

Interpreting Skilled Digital Freelancers as Boundary Spanners

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by

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ABSTRACT OF A DISSERTATION SUBMITTED TO THE FACULTY OF THE
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ABSTRACT

Extant management literature suggests that boundary spanning, defined as linking an organization's internal networks with external sources of information, is crucial to an organization's growth and sustainability in that it enables firms to build connections, share knowledge, and exchange resources. A boundary spanner is a gatekeeper with access to external partners, critical resources, and information, and therefore, plays an important role in creating opportunities for an organization. This dissertation extends the research on individual boundary spanning by examining an overlooked source – skilled digital freelancers – and identifying how and when they are likely to serve as boundary spanners with a primary focus on their boundary spanning behaviors within contracted positions and the facilitating role of human capital and strategic orientation.

In broad terms, this dissertation empirically examines boundary spanning behaviors amongst skilled digital freelancers while considering social embeddedness and structural holes. The findings contribute to a micro-level understanding of organizational boundary spanning in a social network theoretical context. Data for the study consists of pre-piloted survey measures and secondary data gathered on a sample of skilled digital freelancers using the Upwork platform, an online marketplace for connecting freelancers and employers. I conclude with contributions for theory and research and implications for practice.

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I. INTRODUCTION

1.1. INTRODUCTION

1.1.1. Phenomenon of Interest

In today's interconnected, cross-disciplinary, global society, boundary spanning, defined as linking an organization's internal networks with external sources of information, is crucial to an organization's growth and sustainability to build connections, share knowledge, and exchange resources (Dodgson, 1994; Ahuja, 2000; Sandmann, Jordan, Mull, & Valentine, 2014; Mull & Jordan, 2014). However, organizations can struggle to build the collaborative boundary spanning connections on which productivity, efficiency, and innovation increasingly rely (March, 1991; Ernst & Yip, 2009). Organizational and perceptual barriers that limit boundary spanning must be overcome (March, 1991; Clemson, 2012) to permit organizations and their employees to meet the essential need to span boundaries (Ernst and Yip, 2009).

When viewed at the individual level of analysis, a boundary spanner is a gatekeeper with access to external partners, critical resources, and information, and therefore, plays an important role in creating opportunities for an organization (Dodgson, 1994; Ahuja, 2000). Additionally, a boundary spanner is a key employee who serves a dual purpose of processing and diffusing information inside the organization while simultaneously serving as the external face for the organization (Tushman, 1977). Current research emphasizes the benefits of this role for the organization and at times for the individual boundary spanner but is less focused on how someone becomes a boundary

spanner (Soda, Tortoriello & Iorio, 2018). Some noted exceptions have focused on the supervisor's motive to undermine employee boundary spanning (Mell et al., 2021), the role of supportive coaching in encouraging employee boundary spanning (Marrone et al., 2021), and the conceptual influence of employee identification with their organization in determining their engagement with external stakeholders (Korschun, 2015). My dissertation contributes to this less understood micro-level of individual boundary spanning by examining an overlooked individual source – the skilled digital freelancer. Burke and Cowling (2020) demonstrate that freelancers add value to organizations, especially those operating in a dynamic and innovative environment because freelancers improve the ability of the organization to be flexible, innovative, and entrepreneurial. Freelancers are a skilled, specialized workforce that help manage risk in uncertain organizational environments and who “enable businesses to reduce barriers to entry, risk, and financial requirements while enhancing business agility, flexibility, and efficiency. Freelancers are viewed as important inputs that enable an innovation driven and entrepreneurial economy to perform” (Burke and Cowling, 2015: 395). Altman, Kiron, Schwartz, and Jones, (2021) estimates that non-employees (i.e., skilled digital freelancers) perform over 25% of work in an organization and indicate that this is expected to grow.

Skilled digital freelance work involves exchanges and interactions where trust and social relationships connect opportunities across boundaries (Ernst & Yip, 2009; Long, Cunningham, & Braithwaite, 2013). The “definitions of job, employer, and employee are changing, and the complexity of employment relationships requires further examination and reconceptualization” (Kost, Fieseler, & Wong, 2020: 100). Non-

employees, such as skilled freelancers working through digital platforms, perform an increasing amount of the work within organizations (Altman et al., 2021). A skilled digital freelancer is defined herein as someone who works remotely and engages in contract-based jobs for different organizations, often at the same time.

Scholars have begun studying interdependencies that connect intra-organizational relationships with those that go beyond an organization's boundaries (Fuller et al., 2020; Altman et al., 2021). However, application of a boundary spanning lens is surprisingly limited within a skilled digital freelancer context even though, as this dissertation will establish, it is fitting. Theory and findings on external brokers and internal boundary spanners (e.g., employees) does not address the contemporary employment ecosystem that increasingly includes skilled digital freelancers – those with a foot firmly inside and outside an organization, which brings about distinct opportunities and challenges for boundary spanning. Skilled digital freelancers are a distinct and important focus for boundary spanning theory. Boundary spanning theory is specified in relation to traditional employment and brokerage roles – where boundary spanning activities are structured by the specific role and organizational boundaries that are clearly defined. Boundary spanning theory does not speak to 'claiming' the role of a boundary spanner and fluid organizational boundaries, which pertains to the roles of skilled digital freelancers. Research conveys that skilled digital freelancers occupy a unique position (i.e., having one foot outside and one foot inside an organization), which, in this research context, may play a unique role in boundary spanning for a hiring organization and in building the trust needed to have their boundary spanning input received by organizational members. This dissertation examines boundary spanning behaviors,

human capital, and the strategic orientation of skilled digital freelancers and contributes to a micro-level understanding of organizational boundary spanning in a social network theoretical context.

Extant research shows that contractual relationships, like skilled digital freelance work, are complex. It “involves arm’s-length relationships that reduce workers’ responsiveness to the open-ended needs of the employer and distances them from organizational coordination” (Anderson & Bidwell, 2019: 1001). There is research that specifically focuses on the positive and negative outcomes of these positions and relationships within organizations, though not specifically in relation to boundary spanning (e.g., Anderson & Bidwell, 2021; Berti & Pitelis, 2021). Research has been conducted on boundary spanning, social embeddedness theory, and structural holes theory in regard to interpersonal relationships, boundary spanning actions, and relationship quality, though not specifically with regards to freelancers (Huang, Lou, Liu, & Yang, 2016). My dissertation connects these focuses of research.

In addition to the relational aspect of boundary spanning, recent research has focused on structural characteristics of a boundary spanner position within a network, and which behaviors are used to activate the resources that are accessed through their ties to pursue structural opportunities offered by their position in the network (Bensaou, Galunic, & Jonczyk-Sedes, 2014; Kellogg, 2014). The difference and potential complementarity between a social structure and an individual’s boundary spanning capabilities is important to note because networks create space (i.e., the structural opportunity) for social interaction (White, 2008). The developed research model within this dissertation draws on and contributes to this literature by interpreting the behaviors

and strategic orientation of skilled digital freelancers—workers that inherently operate within a structural network—in terms of boundary spanning.

My research framework draws on two social network theories in particular: social embeddedness theory (Granovetter, 1973) and structural holes theory (Burt, 1992). Social embeddedness refers to the extent to which individuals are engaged in stable and repeated relationships (Podolny & Baron, 1997) and shows that individuals make decisions based on past interactions with those they trust (Granovetter, 1985). Structural holes theory furthered social embeddedness theory by defining a structural hole as a gap or missing relationship in a network that prevents information flow between entities that would otherwise be connected (Burt, 1992). In broad terms, this dissertation empirically examines boundary spanning behaviors amongst skilled digital freelancers while considering social embeddedness and structural holes. Extant literature supports the connection of these subtheories to boundary spanning in that boundaries are a basic aspect of organizational life (Ernst & Yip, 2009) and that boundary spanning is associated with bridging structural holes (Schotter, Mudambi, Doz, & Gaur, 2017). This dissertation’s extension of these subtheories to digital freelancers as boundary spanners speaks to the call for scholars to tackle “the challenge of trying to understand the nature of jobs and the management of organizations as they are evolving at the beginning of the 21st century” (Barley, Bechky & Milliken, 2017: 114) and recognizes the “embeddedness of labor in the remote gig economy...is shaped by interpersonal networks of trust generated through micro-level interactions” (Wood et al., 2019: 938).

The gig economy is made-up of platforms that provide access to services by connecting consumers with service providers; services can range from home-repair to

marketing specialists (Fuller et al., 2020). Further, the lack of research using this combined perspective is disconcerting given the tremendous growth of freelance work and the important implications of the boundary spanning process. In this dissertation, boundary is defined as the periphery of an organization (Zobel & Hagedoorn, 2020).

1.1.2. Purpose

The overarching purpose of my dissertation is to examine if and how skilled digital freelancers serve a boundary spanning role within hiring organizations by providing the organization access to external knowledge via the freelancer's network and their capability to bridge structural holes. To do so, this dissertation models and tests primary antecedents to this form of boundary spanning. In particular, skilled digital freelancers' level of interaction with the internal organizational members of a hiring organization is construed as a relevant though not linear antecedent to their boundary spanning. Antecedents pertaining to relational trust and influence also play a central role in the developed model.

A related broad purpose of my dissertation research is the extension of an existing measure of individual boundary spanning behaviors (Ancona & Caldwell, 1992; Marrone et al., 2021) to the domain of skilled digital freelancers. Skilled digital freelancers (e.g., software engineers) construed as boundary spanners aid in connecting external information to the organizations they are working for. I contend that these freelancers coordinate, negotiate, and obtain knowledge, information, and feedback from external stakeholders that is in turn potentially shared with the hiring organization. Therefore, boundary spanning behavior dimensions established in research should generalize to and support the notion that skilled digital freelancers bridge structural holes.

1.1.3. Theoretical, Empirical and Practical Contributions

This dissertation contributes to and extends existing research in three primary ways. First, it extends structural hole theory to the domain of skilled digital freelancers. Though scholars have applied this theory to examine agent, broker, and other external bridging roles (e.g., Burt, 2007; Kwon et al., 2020; Lin et al., 2021), these individuals and entities reside firmly outside of the organization. A skilled digital freelancer occupies the unique position of someone that is simultaneously external and internal of the organization (Ernst & Yip, 2009; Long, Cunningham, & Braithwaite, 2013). To extrapolate to the freelancer context, I integrate relevant concepts and findings from the internal organizational boundary spanning literature and particularly regarding organizational teams. Together, this offers not only an empirical extension of structural hole theory to a new domain by studying a unique sample consisting of freelancers, but also a theoretical explication of its potential linkage to intraorganizational dynamics by examining if and how freelancers serve as boundary spanners.

Second, this research contributes to an understanding of specific individual human capital components like education and working experience in facilitating individual boundary spanning behaviors. This is a particularly novel view within boundary spanning research and applies to externally positioned boundary spanners. I frame individual experience as an antecedent to freelancer's ability to contribute new information to the hiring organization and as a factor in freelancer's influence over organizational members to accept their input. Boundary spanning in this light essentially translates the more structurally oriented view of boundary spanning (Burt, 1977) to the

individual perspective and permits a more self-directed way of viewing boundary spanning.

Third, this dissertation addresses a gap in the theoretical and empirical understanding of how individuals become boundary spanners. There is a consensus that boundary spanning positions provide opportunities to individuals (Burt, 1997). However, research on their motivation and intent is not fully understood (Obstfeld, Borgatti, & Davis, 2014; Stovel & Shaw, 2012). The following applies trust and related power dynamics as theoretical underpinnings for predicting skilled freelancer's boundary spanning behaviors towards their hiring organizations. I consider the individual strategic orientation of skilled freelancer's, collaborate versus arbitrage, and freelancer ability to influence internal organizational members, both to probe motive and perceived intent as antecedents to boundary spanning behaviors.

The findings also hold potential implications for practice. For instance, hiring organizations are increasingly building their workforce to include ongoing use of skilled freelancers (Fuller et al., 2020) and are hastening their technological processes to connect with external parties (Altman, Nagle, & Tushman, 2015). However, the strategic intent to gain boundary spanning advantages is not necessarily explicit in this growing approach. At the same time, extant management literature suggests that boundary spanning on both an internal and external level is central to organizational growth and success (Williams, 2002) and scholars acknowledge its relevance in management (Rosenkopf & Nerkar, 2001).

1.1.4. Dissertation Format

This dissertation is structured as a single-study research monograph. The outline of the dissertation chapters following this introduction chapter are as follows: First, a theoretical overview is provided on boundary spanning, social embeddedness theory, and structural holes theory, ultimately linked together for a presented theoretical framework. Then, hypotheses are developed to examine skilled digital freelancers access, ability, and motive as antecedents to their boundary spanning behaviors. Next, methods are presented for both a pilot and main study with a focus on the main study.

II. THEORETICAL DEVELOPMENT

2.1. THEORETICAL DEVELOPMENT

2.1.1. Background

Boundary spanning examined at the individual level fundamentally represents the interpersonal business relationships developed across organizational boundaries. Extant research shows this typically takes the form of relationships that a select set of organizational employees has with external stakeholders (e.g., Huang et al., 2016; Korschun, 2015). In other words, a boundary spanner's relationships go beyond intraorganizational relationships with co-workers and includes relationships with external members like buyers, suppliers, and members of other organizations (Tushman, 1977). This is an inside-outside organizational view of boundary spanning relationships. My research builds on and extends research in the boundary spanning domain by modeling an outside-inside organizational interpretation of boundary spanning enacted by skilled digital freelancers who are not bound to a single organization due to the nature of their employment relationship.

Specifically, I focus on the boundary spanning behaviors of skilled digital freelancers as quasi organizational members in building business relationships with internal organizational members. The overarching objective of this dissertation is to explicate boundary spanning behaviors that skilled digital freelancers exhibit when interacting with internal organizational members, primary antecedents to these behaviors, and how organizational members are influenced to accept the boundary spanning input of

skilled freelancers. My framework is grounded in two social network sub-theories known as structural holes (Burt, 1992; 1997) and social embeddedness (Granovetter, 1973), and developed through a boundary spanning lens (Aldrich & Herker, 1977).

Boundary spanning research encompasses a substantial variety of labels. Brokers and boundary spanners are two of the most common descriptors in a lengthy list of synonyms or closely related constructs representing roles that reflect a connectivity function across a structural hole (Long et al., 2013). Extant strategic management literature often uses the terminology ‘broker,’ an “individual positioned between unconnected others” (Soda, Tortoriello & Iorio, 2018: 896). Brokers are generally viewed as external of the organization, like skilled digital freelancers, and do not have a defined internal role. The role of a broker has been researched by organizational and social network scholars interested in uncovering how network advantages are distributed in social structures. For purposes of this dissertation, the terms boundary spanner and boundary spanning are used to refer to brokers in the broadest sense.

2.1.2. Boundary Spanning Lens

Boundary spanning as an organizational function involves building relationships with internal and external stakeholders (Mull & Jordan, 2014) and leveraging expertise to solve problems (Sandmann et al., 2014). Early literature established that boundary spanning serves two main functions: information processing and external representation (Aldrich & Herker, 1977). Through these functions, boundary spanning supports the exchange of information with the external environment and manages organizational responses to environmental influences and exogenous shocks (Tushman & Scanlan, 1981) like the 2020 COVID-19 pandemic. Therefore, knowledge sharing through

boundary spanning aids organizations in learning new knowledge, building relationships, and developing novel innovations (Huang et al., 2016; Iorio, 2022).

Boundary spanners as individuals embody a diverse set of behaviors and perform various activities that are necessary not only to build support for new products and projects, but also to coordinate innovation efforts across organizational boundaries and groups (Ancona & Caldwell, 1990). “Boundary spanners facilitate transactions and the flow of information between people or groups who either have no physical or cognitive access to one another, or alternatively, who have no basis on which to trust each other” according to one review of boundary spanning articles to determine how boundary spanners are identified (Long et al., 2013: 1). They are individuals who network, build relationships and trust (Schotter et al., 2017), connect organizations with new, external sources of information (Mull & Jordan, 2014; Tushman, 1977), and connect opportunities within and across networks (Ireland & Webb, 2007; Granovetter, 1973, 1985; Burt, 1992) by offering support and information for organizational growth and development (Ireland & Webb, 2007). Additionally, they seek out novel ideas (Aldrich & Herker, 1977), communicate with members of various networks (Friedman & Podolny, 1992), and assist organizations with innovation due to inherent knowledge and collaboration efforts (Koufteros, Vonderembse, & Doll, 2001).

The behaviors described above reflect that boundary spanners tend to engage in activities on both sides of the periphery of an organization, which implicitly requires some degree of permeability. Zobel and Hagedoorn (2020: 400) found that “firms need to increase the permeability of their organizational boundaries to enable knowledge exchange with a broad set of partners.” It is the permeability of boundaries that enable a

balance between internal and external knowledge (Casper et al., 2018; Wayne et al., 2017). Organizations achieve novel innovations by allowing “inflows and outflows of knowledge to accelerate internal innovation and expand the markets for external use of innovation” (Chesbrough, Vanhaverbeke & West, 2006: 1). This is important for organizational growth and sustainability because it allows knowledge sharing across boundaries (Chesbrough et al., 2006). Despite this recognized importance, research on permeable organizational boundaries is limited in its micro-level understanding of how individuals actually span the periphery, which is where this dissertation’s focus on boundary spanning individuals in the form of digital freelancers comes into play.

2.1.3. Social Network Theory

Social network theory describes the relationships among individuals, groups, and organizations and how they interact, emphasizing the mechanisms and procedures used to obtain outcomes like having numerous ties and connections or being centrally located within a network (Brass, 2002). Social network theory views relationships in terms of nodes and ties where nodes are the individuals within the network and ties are the relationships between the individuals. Therefore, a social network is simply a map of relevant ties between contacts. Social network theory states that ties within and outside a network are important because they can create or constrain information flow between contacts (Borgatti & Halgin, 2011).

In extant literature, social networks have been used to examine how organizations interact with each other and to characterize the many informal ties that link employees together like the associations and connections between individuals at different organizations (Soltis, Brass & Lepak, 2018). These networks provide ways for

organizations to gather information, deter competition, and define strategies. Scholars have stated that network theory should be used as the function of enabling something to flow from one node to another (Borgatti & Halgin, 2011). Two social network sub-theories that focus on distinct aspects of how networks operate and act as paths of information flow and knowledge sharing are social embeddedness theory (Granovetter, 1973) and structural holes theory (Burt, 1992). The relevant broad aspects of each subtheory for the context of this dissertation are described next.

2.1.3.1. Social Embeddedness Theory

Social embeddedness refers to the extent to which individuals are engaged in stable and repeated relationships (Podolny & Baron, 1997) and shows that individuals make decisions based on past interactions with those they trust (Granovetter, 1985). The composition of networks and relationships may impact an individual's ability to span boundaries (Brion et al., 2012). For example, "Personal skills and background cannot be considered in isolation, as to be effective they must be associated with a suitable personal network" (Brion et al., 2012: 709). Granovetter (1985) applied embeddedness to market societies which showed that economic exchanges are influenced by pre-existing social ties. Therefore, work exchanges may become embedded within networks (Granovetter, 2005) which happens through micro-level interaction and can lead to the development of interpersonal trust. Developing trustworthy relationships is imperative and is based on communication, interaction, and perception (Granovetter, 1985).

Polanyi (1957) argued that more than being embedded, organizations and individuals must be understood as part of a larger social structure to enable knowledge and resource sharing. Individuals involved in making business decisions and performing

operations are more likely to be affected by their relationships as they tend to work with others more closely when compared to those who do not make business decisions or perform business operations (Huang et al., 2016). For example, the actions they choose can change by the social relationships they function in and are embedded. The key notion is that individuals perform their actions by taking the choices of others into account, meaning, their social ties are critical to their actions due to social influences (Granovetter, 1985). Embeddedness organized on the foundation of solid relationships in particular plays an important part in the establishment of a relational code of conduct, mutual awareness, and cooperation (Uzzi, 1996; Rowley, Behrens, & Krackhardt, 2000). Social embeddedness views employees as organizational representatives who symbolize the values of the organization, job, and work (Hayton, Carnabuci & Eisenberger, 2012).

By extending social embeddedness logic, extant literature argues that transactions and relationships are embedded in the relational ties of boundary spanners (Huang et al., 2016). Consistent with social embeddedness theory (Aldrich & Herker, 1977), boundary spanners maintain relationships and interact both outside and within their social networks. Boundary spanners are considered key players, meaning, their loss from a network could greatly affect its functionality (Long et al., 2013). Huang et al. (2016) draws on social embeddedness theory to propose that the strong ties boundary spanners establish may benefit exchange parties in their relationships; for example, skilled freelancers and hiring organizations. Boundary spanners do so to build trust, perform tasks, and achieve goals (Ernst & Yip, 2009).

However, Borgatti and Halgin (2011: 1172) propose that “nodes that are embedded in locally dense parts of a network will often receive the same bits of flow

from their various contacts, because the contacts are tied to each other.” This reflects the limitation of so-called strong ties, that the stronger the connection between two entities, the more likely their social networks overlap, therefore limiting new information due to those within that network sharing the same relationships and knowledge (Granovetter, 1973). Conversely, the strength of weak ties (Granovetter, 1973) explains why some individuals have access to more and better opportunities than others. Weak ties refer to the connections between individuals who do not share the same relationships and who are a part of other distinct networks. Granovetter (1973) argued that there is strength in weak ties like gaining access to and being better connected to a broader network. For example, individuals typically gain job offers through acquaintances rather than strong relationships. Granovetter (1973) conducted a study of 200 people in new jobs and asked how they secured them. Approximately 75% said they secured their jobs from acquaintances, also known as weak ties.

Individuals with more weak ties have a relatively greater diversity of connections and access to a variety of information which helps obtain more insight into the needs and capabilities of the various networks to which they are connected (Granovetter, 1973). Individuals with weak ties are better able to link information, develop innovations, and generate ideas (Burt, 2004; Tortoriello, 2015; Goldberg et al., 2016). Weak ties are a significant correlate of novel ideas because they encompass more diversity and non-redundant information (Burt, 2004; Zaheer & Soda, 2009). The core concept is that weak ties can be more important than strong ties in terms of providing novel and actionable information, like the weak tie between a skilled freelancer and a hiring organization.

In a recent study by Rhee and Leonardi (2018), employees with more weak ties or less-constrained networks produced better ideas than those with strong ties or constrained networks. When two separate individuals possess non-redundant information, there is a structural hole between them, meaning a gap between that relationship or connection (Burt, 1992). Weak ties can form bridges of new information between networks that might otherwise never interact, therefore, weak ties can act as bridges between structural holes, which in turn is shown to significantly impact an individual's innovation (Iorio, 2022). To put it simply, bridging ties are a potential source of generating novel ideas and gathering new knowledge, therefore, weak ties are likely to be bridges, like the framework in this study suggests.

2.1.3.2. Structural Holes Theory

Structural holes theory furthered social embeddedness theory by defining a structural hole as a gap or missing relationship in a network that prevents information flow between entities that would otherwise be connected (Burt, 1992). It is a divide in a network and a disconnection between entities, meaning, a lack of connection through which information should be flowing (Burt, 1992). A structural hole is considered “an open triad, an open structure, a lack of closure, or a lack of network constraint” (Kwon et al., 2020). Whereas Granovetter (1973) emphasized the strength of a tie (strong versus weak) in determining whether it plays a bridging role, Burt (1992) focuses more directly on the structural hole itself.

As social network theory states, “ties interconnect through shared end points to form paths that indirectly link nodes that are not directly connected. The pattern of ties in a network yields a particular structure, and nodes occupy positions within this structure”

(Borgatti & Halgin, 2011: 1169). Yet such networks contain disconnects or structural holes that hinder diffusion of information. Those who act as bridges are crucial in maintaining the diffusion of information. They attain advantageous positions like gaining access to opportunities since they control the flow of information between diverse relationships, create information benefits for the network and for themselves, and acquire more information and opportunities than those who do not bridge structural holes (Burt, 1992; 2007; Soda et al., 2018). However, in addition to increasing opportunities, a structural hole decreases trust (Ahuja, 2000). This fundamental point is consistent with the strength of weak ties logic elaborated within the social embeddedness perspective (Granovetter, 1973).

Structural holes theory perceives information exchange as a mechanism that turns relationships into assets (Coleman, 1988; Burt, 1992). For example, a “structural hole is an opportunity to broker the flow of information between people and to control the form of projects that bring together people from opposite sides of the hole” (Burt, 1997: 340). Interactions and opportunities created by structural holes determine how networks are formed. Past network connections aid individuals due to previous experiences and knowledge which creates future structural hole opportunities (Zaheer & Soda, 2009). In recent literature, Lin et al. (2021) defined the term ‘structural hole spanner’ as an individual who bridges structural holes. The study found that structural hole spanners occupy crucial positions which connect various relationships and play a key role in spreading information in social networks. The study considered the impact of these spanners on social networks through information and control advantages and claimed it is

essential to detect these individuals within social networks, a point to which this dissertation research speaks.

2.1.4. Connecting Social Network Theory, Boundary Spanning, and Freelancers

Organizational social networks should be diverse to gain knowledge, contribute to the sustainability of the organization and its network, and to bridge structural holes (Burt, 2007; Martínez-Pérez, Elche, & Garcia Villaverde, 2019). The proposed research framework explores whether skilled digital freelancers are the gatekeepers who occupy structural hole positions and act as boundary spanners to bridge structural holes. Skilled digital freelancers are a potentially fluid and efficient way for organizations to bridge structural holes and remains unexamined within micro-level boundary spanning research, which typically focuses on intraorganizational members (e.g., Marrone et al., 2021; Mell, et al., 2022; Iorio, 2022) or external brokers (e.g., Soda, et al., 2018).

However, the subtheories of social embeddedness and structural holes point to a potential tension in the position of freelancers as boundary spanners. Whereas weak ties facilitate sources of new information, it affords lower relational trust relative to strong ties (Iorio, 2022). Lower trust may affect freelancers' willingness to share discretionary information and may affect organizational members willingness to accept freelancers sharing of information. For example, according to Soda et al. (2018: 899), arbitraging versus collaborative brokers “are more likely to strive to access, control, and use others’ knowledge and information in a proprietary fashion.” In this way, they try to maximize the attainment of appropriate knowledge and information gathered through their network of contacts, do not share this knowledge and information openly with others, and attempt to individually benefit. This type of orientation also occurs with skilled digital freelancers

who actively seek and find diverse knowledge bases (e.g., leveraging ties and spanning across multiple disconnected organizations, industries, and markets), which are then gathered, combined, and integrated into novel ideas like solutions and innovations (Hargadon, 2003).

Knowledge opportunities available through freelancers' networks can be a motivating factor to bridge ties (Lingo & O'Mahoney, 2010). However, due to the transient environment of skilled digital freelance work, there is limited time to develop trust (Nath, 2020). Trust is "the willingness of a party to be vulnerable to the actions of another party based on the expectation that the other will perform a particular action important to the trustor, irrespective of the ability to monitor or control that other part" (Mayer, Davis, & Schoorman, 1995: 712). Mayer et al. (1995) states that trustworthiness should be understood as a continuum. Based on this perceived continuum, elements of boundary spanning may come into existence. For example, the relationship between a skilled digital freelancer and hiring organization having more trust may encourage the skilled digital freelancer to engage in a collaborating strategic orientation with boundary spanning behaviors. Indeed, a recent study demonstrated that boundary spanners' level of collaboration in a network varies independently from the network's structural characteristics (Soda, Stea & Pedersen, 2019).

Extant research has probed trust in relation to boundary spanning from different perspectives, finding for example that being perceived as a boundary spanner could create negative perceptions (Evans et al., 2015) because boundary spanners can use their role in a self-serving way. (Parkinson, Kleinbaum, & Wheatley, 2017). Mell et al. (2022) found that boundary spanning can be perceived as an act of undermining because it

weakens a supervisor's territorial control over their group. However, if a boundary spanner seeks advice from their supervisor first, it could decrease the negative interpretation of their motive and intent (Mell et al., 2022). Additionally, Iorio (2022) examined that while bridging structural holes and utilizing weak ties provided information and timing benefits, it was perceived as being self-serving and decreased a boundary spanners reputation and relational trust. Iorio (2022) labels this as actual versus perceptual brokerage; actual refers to weak ties and perceptual is essentially how trust is managed.

The hypotheses chapter probes this point of tension surrounding weak ties and trust within boundary spanning by examining freelancer boundary spanning behaviors as conditioned on their interaction with organizational members, their ability to use expert power in influencing organizational members, and the degree of their collaborative orientation. More holistically, this dissertation framework examines skilled digital freelancers access, ability, and motive as antecedents to their boundary spanning behaviors. In doing so, this proposed study also speaks to how the two main facets of network advantages, boundary spanning as a structural position occupied in a network and boundary spanning as a strategy for networking actions (Obstfeld et al., 2014), applies to skilled digital freelancers. The comprehensive integrated conceptual depiction of freelancers is displayed in Figure 1 and the empirical model is shown in Figure 2.

Insert Figure 1 about here

Insert Figure 2 about here

III. HYPOTHESIS DEVELOPMENT

3.1. HYPOTHESIS DEVELOPMENT

3.1.1. Antecedents

To empirically examine the construal of freelancers as boundary spanners, I focus on theoretically necessary antecedents to their boundary spanning behaviors: 1) freelancer interaction with internal organization members, 2) degree of freelancer human capital along with the intervening variable of freelancer expert influence, and 3) freelancer strategic orientation for collaboration versus arbitrage. These antecedents represent, in order, freelancer's *access*, *ability* – including human capital and “expert power”, and *motive* for boundary spanning.

First, regarding freelancer access, I extend known manifestations of boundary spanning to the domain of freelancers by construing which types of boundary spanning behaviors generalize to freelancers. I assess boundary spanning behaviors from the perspective of the freelancer's role when working for organizations. In this regard, I suggest that freelancer boundary spanning behaviors include not only their contracted duties, but also what they informally choose to bring to the role in terms of discretionary expertise. This is consistent with prior research on boundary spanning behaviors, as subsequently discussed, that tends to focus on traditional employee roles and behaviors that are discretionary or non-explicit. Building on this fundamental point, I then delineate the necessity of the freelancer's interaction with other members in the organization. For example, a position within a team makes it possible for a freelancer to convey their

externally gained knowledge, expertise, and information both formally and informally to internal members of the organization, whereas solely independent work may bound or limit the exchange. As I build the conceptual rationale for hypothesis 1 the subtlety of this perspective will become apparent.

Next, regarding ability, I examine individual differences in the professional capabilities of freelancers as an antecedent to their propensity to perform boundary spanning behaviors. I focus on the human capital attributes that make it possible and more likely for a freelancer to bridge structural holes in the hiring organization's knowledge, expertise, and information. The basic premise is that freelancers with higher levels of human capital are more likely to have the objective *and* perceived capability to contribute resources that the organization's existing workforce does not possess, versus simply contributing routine labor. For example, a freelance coder with relatively little experience or training is more likely to serve (and be viewed) as 'just' another set of hands performing lower-level functionary work as the organization expands and contracts its workforce to meet demand. Conversely, a freelance coder with a relatively high-level of expertise has greater potential to contribute new information and capabilities that serve to bridge a gap in the company's existing structure and is more likely to have the influence in the form of accepted expert power to do so. Therefore, the mediating role of expert power is examined when considering the relationship between freelancer human capital attributes and boundary spanning behaviors, suggesting that human capital may lead to increased expert power, which, in turn, facilitates boundary spanning. The development of hypothesis 2 builds this argument more fully.

Finally, regarding motive, I turn to the freelancer's individual strategic orientation for collaboration versus arbitrage as a moderating effect between their expert power and level of activity in boundary spanning behaviors. While individual strategic orientation can have a dispositional component, my implicit focus is on the presence or absence of trust between freelancers and hiring organizations. The development of hypothesis 3 discusses trust and related power dynamics as theoretical underpinnings for freelancers' strategic orientation towards their role with hiring organizations.

3.1.2. Hypothesis 1

The organizational view of boundary spanning, as outlined in the theoretical overview, encompasses a vast set of behaviors and activities. However, there are some established boundary spanning behaviors that are conceptually unique and apply only to certain actors. For instance, behaviors that span the organization's internal hierarchy (Joshi, Pandey, & Han, 2009), its internal functional departments (Whelan & Teigland, 2013), or that require navigation of internal power dynamics to connect internal sources of attention and resources (Goldring & Sims, 2005) are typically out of reach of most freelancers due to their limited internal footing within an employing organization. Conversely, boundary spanning behaviors that reside predominantly outside of the organization do not capture the unique position of the freelancer as someone with a presence both within and outside the organization. For instance, certain external agents, as boundary spanners, rely exclusively on connecting organizational members to an external stakeholder or network, such as coordinating with stakeholders, pushing for management support (Marrone, 2010; Sheremata, 2000), or promoting external

representation (Aldrich & Herker, 1977); however, these do not capture the unique dual position of a skilled freelancer.

The unique outside-inside position of the freelancer is well suited to bridging gaps in knowledge, information, and skills that are lacking, to at least some extent, within the organization. Stated in terms of structural holes, people that are uniquely connected across different groups gather and carry alternative ways of thinking across the structural holes that exist between such groups (Burt, 2004). A review of boundary spanning research outside the field of management similarly shows the role of boundary spanning as supporting the transfer of otherwise isolated knowledge and information (Long et al., 2013). Freelancers, in particular, use networks to develop the knowledge and skills they need to find higher paying jobs and to navigate the competitive gig economy (Aguinis & Kraiger, 2009). Schwartz (2018) discovered that freelancers who actively used job-related communities received valuable feedback and appraisal on projects, which helped them further develop important competencies. With this knowledge and skills, freelancers can broker the exchange of information from their network to the organization. It is in this way, I suggest, that freelancers may possess the information needed to bridge an organizationally relevant structural hole.

Regarding the other side of the structural hole (i.e., how freelancers bring this information into the organization), I reason freelancers bring this information into the organization when they are engaging with the relatively isolated internal employees at the organization's work team level. For instance, freelancers are known to work as members of project teams (Ratcheva, 2009) and are commonly hired to work in conjunction with or under the supervision of existing organizational employees rather than in independent

isolation (Brion et al., 2012). Considering freelancers from this perspective, their boundary spanning actions conceptually align with extant team-focused boundary spanning behaviors labeled in prior research as scouting, which encompasses general information search and scanning of the external environment to gain access to relevant information, knowledge, and expertise (Ancona et al., 1992). Further, freelancers based on the external nature of their position have little common ground with other known boundary spanning behaviors at the internal work team level, which include coordinating with stakeholders and lobbying for management support and resources (Ancona & Caldwell, 1992; Marrone, 2010; Sheremata, 2000). Research on contractor-managers operating at a group level suggests a low likelihood of involvement in political organizational aspects relative to internal organizational members (Anderson & Bidwell, 2019).

Thus far, I have argued that structured interaction between the freelancer and organizational members, such as and in particular when a freelancer works as part of an internal project team, facilitates freelancer sharing of information, knowledge, and expertise to fill recognized gaps among organizational members. However, it could be that freelancer boundary spanning behaviors also occur outside of structured team settings and in a less interactive way. Boundary spanning research on brokers suggests that very little or no interdependence is required for boundary spanning benefits to occur (Soda et al., 2018). More aligned with the present research context, recent research on contractor-managers indirectly suggests that boundary spanning could occur at a dyadic level. Anderson and Cappelli (2021) found that external experience gave contractor-managers an important source of expert power when working with both individuals and

teams, or an ability to influence organizational members with their expertise regardless of degree of formalized interaction and interdependence. Implicit in this is that some level of interaction occurs, but that boundary spanning benefits could effectively occur at lower degrees of interaction than suggested by membership in a project team.

Taken together, I contend that freelancer interaction with internal organizational members plays a positive role in freelancer's boundary spanning behaviors but establishing at least some interaction may be incrementally more impactful than establishing more. Whereas existing research on traditional boundary spanners supports a case for their interaction at each side of a structural hole, it does not specifically address degree of interaction. For example, internal boundary spanners are connected to information external to the organization, which aids in processing and disseminating information internally (Whelan et al., 2013), and communication is important as it relates to transferring external sources of information to within the organization (Tushman & Scanlan, 1981). Extending these same characteristics to skilled freelancers would potentially require that they interact with internal organizational members only to the extent necessary to transmit external resources. All considered, hypothesis 1 poses a positive curvilinear relationship between a freelancer's degree of interaction with internal organizational members and a freelancer's boundary spanning behaviors to reflect interaction as a necessary condition but one with a potential saturation point, meaning, more interaction does not equal commensurately more sharing in a curvilinear relationship; there are diminishing returns.

H1: Skilled freelancer degree of interaction with internal organization members has a positive curvilinear relationship with their degree of boundary spanning behaviors performed in contracted roles.

3.1.3. Hypothesis 2

As described in the development of hypothesis 1, the primary way in which freelancers span boundaries in their role for a given organization is by filling the knowledge gaps—transferring external knowledge gained to within the organization. Tushman et al. (1981) claim organizations need human capital that can span boundaries, gain knowledge, and transfer that knowledge to other groups and domains. Skilled freelancers potentially fulfill this need based on my earlier reasoning. However, even among skilled freelancers, the degree of skill or human capital one holds varies. Existing literature suggests that examining the nature of human capital among boundary spanners is imperative (Wright et al., 2007).

Human capital comprises the wealth of knowledge and skills that exist within individuals (Becker, 1964). Education and experience are both traditional measures of human capital, and certificates completed, and working history are common objective ways they translate in the skilled digital freelance environment. For example, skilled digital freelancers must portray the quality of their work, credentials, and skills in a clear way (Droege, Anderson, & Bowler, 2003). A recent study found that 33% of freelancers believe that certifications aid in their career advancement, in areas such as marketing, networking, and software development (Warner, 2020). Freelancers with these kinds of credentials are considered more capable and preferred (Warner, 2020).

Several studies share a common human capital assumption; individuals with more human capital, like education and experience, are more productive than comparable individuals due to their ability to solve problems, adapt to changing environments, and contribute to a firm's strategic capabilities (Becker & Huselid, 2006; Coff & Kryscynski,

2011). All else equal, a higher level of human capital suggests a greater potential to contribute resources that the organization's existing workforce does not possess, making it possible for a freelancer to bridge structural holes in the hiring organization's knowledge, expertise, and information. Indeed, Kazi and Yusoff (2014) claim that human capital is likely more important to a skilled freelancer than a typical employee. This reasoning provides the basis for hypothesis 2a, that freelancer level of human capital has a positive relationship with their degree of boundary spanning behaviors.

However, the underpinnings of how human capital affects boundary spanning remains unexplored. To flesh out the intervening logic that connects freelancer human capital with their boundary spanning behaviors, I extrapolate from how human capital has been applied to a manager-contractor setting. Anderson et al. (2021) studied managers in temporary, contract-based roles and found that despite being faced with challenges, most are successful. This success is primarily due to a specific set of behaviors they exhibit and utilize, that differ from the more typical managerial behaviors of planning and directing. During interviews, managers in temporary, contract-based roles said they relied on their expertise and credibility from past experiences more than in previous traditional employee-manager roles (Anderson et al., 2021).

Human capital differs amongst employees and creates power shifts (Tzabbar, 2009). For example, employees with expertise gain more power and an influential status when compared to employees with less expertise (Kehoe et al., 2015); this is known as expert power (French et al., 1959). In the context of freelancers, expert power is based on employee's perception that a freelancer has specialized knowledge that the organization does not currently possess. Anderson et al. (2021) findings suggest that freelancers

should remain independent of the organization to better leverage their expertise rather than establish themselves as part of the organization.

In sum, extant literature suggests that boundary spanning involves expert power, meaning, a freelancer's influence over internal members to accept their ideas (Anderson et al., 2021). Taken together, the causal argument here is that freelancer degree of human capital, such as level of education and experience, signals greater expertise that, in turn, may make organizational members more receptive to and even seeking out of the freelancer's ability to fill a structural hole. Therefore, I argue that skilled freelancers with higher levels of human capital will perform higher levels of boundary spanning, *ceteris paribus* (controlling for degree of interaction with internal members), and that expert power will mediate that relationship. I hypothesize:

H2a: Skilled freelancer degree of human capital has a positive relationship with their degree of boundary spanning behaviors performed in contracted roles.

H2b: The relationship between skilled freelancer degree of human capital and degree of boundary spanning behaviors is mediated by their subjective level of expert power.

3.1.4. Hypothesis 3

Whereas hypothesis 1 and 2 focused on freelancer access and ability for boundary spanning, hypothesis 3 will focus on the effect of freelancer motive on their boundary spanning behaviors based on findings that individuals may strategically limit sharing of knowledge, expertise, or connections to preserve their expert position (Kehoe & Tzabbar, 2015; Issac et al., 2022; Soda et al., 2018). Specifically, I consider individual strategic orientation towards collaboration versus arbitrage regarding a skilled freelancer, while

considering a lens of trust between skilled freelancers and hiring organizations. From a theoretical network perspective, trust aids in collaboration (Jarvenpaa & Shaw, 1998).

Conceptual research on the productivity of work teams comprised of internal organizational members and external contributors emphasizes the role of individual self-interest in determining cooperation and contribution to the organization (Berti & Pitelis, 2021). Individual and collective interests may misalign, in keeping with the notion of “wide variance in the extent to which individuals benefit from bridging structural holes” (Burt, 2012: 587). In research focused on external brokers (a form of boundary spanner), Soda et al. (2018) empirically supported two strategic orientations of this type of boundary spanner: arbitrage and collaboration. An arbitraging strategy exploits information asymmetries offered by the boundary spanner’s structural position among disconnected others, whereas a collaborating strategy connects disconnected others in a way that favors open and complete knowledge sharing among all involved parties (Soda et al., 2018). According to Soda and colleagues (2018: 899), an arbitraging broker is “more likely to strive to access, control, and use others’ knowledge and information in a proprietary fashion.” In this way, they try to maximize the attainment of appropriate knowledge and information gathered through their network of contacts, share it selectively, and attempt to individually benefit from its integration.

In considering whether skilled freelancers might vary in degree of collaboration or arbitrage within hired roles, it is useful to consider the role of trust towards the hiring organization. Trust aids in forming a relationship that is open to knowledge sharing and information exchange and is often based on past experiences (Lipnack & Stamps, 2008). Similar to the study from Soda et al. (2018), some freelancers may not be willing to share

beyond their specific contracted organizational role due to a lack of trust, and instead may choose to arbitrage and withhold additional information for strategic reasons such as wanting to remain needed, leverage power, or for some other self-interested reason (Burt & Wang, 2021). For instance, Kehoe et al. (2015) findings demonstrate that a desire to preserve expert power can limit knowledge sharing by star employees. Thus, a skilled freelancer may become more reserved with their boundary spanning behaviors if trust is lacking or if self-interest is dominant and may not share the knowledge, skills, and power they have – at least at the discretionary level beyond what is required for the contracted role and to manage their reputation rating.

Beyond the motives of trust and related self-interest, contextual variables may also precede orientation in that freelancer-project fit, organizational environment, and supervisor support are shown to positively influence project engagement of the skilled freelancer and, in turn, their degree of knowledge sharing (Barlage, van den Born, & van Witteloostuijn, 2019). Together, these findings support the potential for variation among freelancers in their strategic orientation, and therefore propensity toward knowledge sharing or boundary spanning behaviors despite ability to do so. Building upon hypothesis 2b, this forms the basis to pose a moderated mediation relationship for hypothesis 3:

H3: Freelancer collaborate orientation amplifies the positive relationship between freelancer expert power and their degree of boundary spanning behaviors.

IV. METHODOLOGY

4.1. METHODOLOGY

4.1.1. Research Context

4.1.1.1. Skilled Digital Freelancers

The working environment within the gig economy indicates “a way of working that is based on people having temporary jobs or doing separate pieces of work, each paid independently, rather than working for an employer” (Cambridge Dictionary, 2020).

Skilled digital freelancers are individuals who find work via digital labor platforms like Upwork. For purposes of this study, a skilled digital freelancer is defined as an individual who works on several projects at the same time or within a time frame, but for different organizations (Anderson et al., 2019), and who provide a skilled service or labor, and not those who rent out assets (e.g., Airbnb), sell goods (e.g., Etsy), or perform unskilled labor (e.g., Uber driver) (Kuhn & Galloway, 2019). Therefore, this study focuses on professional and skilled freelancers like accountants and software engineers that get hired through Upwork and work for organizations, rather than other types of contractual workers (e.g., Uber Drivers). It is important to understand that my research does not study the unskilled labor market gig economy but focuses on skilled freelancer professionals.

4.1.1.2. About Upwork

Upwork and other skilled freelancer platforms (i.e., fiverr.com or freelancer.com) are technology platforms that organize and sell the labor of knowledge workers. Due to

the focus on skilled professionals, extant research utilizes terms like ‘knowledge work’ or the ‘knowledge economy’ (Litchfield, Hirst & Daan van Knippenberg, 2021). Upwork provides an online platform for organizations (also called clients) in need of specific professional services to connect with skilled freelance workers. Upwork has become the world’s largest online platform with \$2.5 billion of transactions in 2021, over 16 million skilled freelancers, and 5 million clients from over 180 countries (Zhou, 2022).

One in five Fortune 500 companies hire skilled freelancers through Upwork portraying that there are tremendous opportunities for both skilled freelancers and organizations who utilize the platform (Upwork.com, 2021). I selected Upwork for my study because it provides the ability for organizations to interview, hire, and work with skilled freelancers from various industries and skill sets. Additionally, skilled professionals, like accountants and software engineers, can bid out their services to obtain jobs. Upwork allows a skilled freelancer to choose and apply for jobs and receive revenue earned for each job, and organizations can hire and pay freelancers (Kallenburg & Dunn, 2016).

The information on a freelancers Upwork profile includes skills, a list of previous jobs with associated hourly wages, education, and earned certificates (Upwork.com, 2021). Job postings from clients contain information about the job and a list of required skills (Upwork.com, 2021). Job opportunities on Upwork range from small tasks like data entry and software testing, to larger-scale projects like application development and website design.

Online freelance labor is a growing part of the economy and is driven by a combination of better technology and evolving attitudes toward career change. Research

trends indicate that freelancer growth will continue. As a result, extant research has utilized Upwork as a valid form of data collection and analysis (Seifried, 2018; Zaggl & Aarhus, 2020). A sample of Upwork freelancers is a natural data source for this study and generalizable to other freelancers who operate outside of the platform because they operate in a similar manner by securing jobs at various organizations and with a multitude of clients. Additionally, freelancers' function by having a foot both inside and outside a given organization. For my study, I will collect, utilize, and analyze data from two distinct samples of skilled freelancers via Upwork, which is discussed in sample and data collection.

4.1.1.3. Sample and Data Collection

My study investigates the boundary spanning behaviors performed in contracted roles by comparing two distinct sample groups of skilled digital freelancers attained through Upwork. Through purposeful sampling, the two sample groups differ in their likely degree of interaction with internal organization members, which is a relevant aspect of the developed model. Sample one is freelancers who are more team focused, (software engineers). Sample two are freelancers who work more independently (accountants). This purposive sampling objectively captures structural variation in freelancer interaction with organizational members. Further, while these two sample groups are distinct in this variable of interest, they remain similar in other potentially confounding aspects because both groups are composed of skilled freelancers currently employed via Upwork and based within the United States.

My study utilizes a survey instrument that was administered to the Upwork sample to measure boundary spanning behaviors, subjective expert power, strategic

orientation, and a scaled measure of interaction with organizational members to supplement the variation captured through purposive sampling. In addition, secondary data was gathered in the form of human capital measures reported on the freelancer platform profile and within their resume, and controls as described subsequently. This secondary data permits measurement of the independent and dependent variables from separate sources. Avoidance of potential same-source bias in measurement is also supported by the two sample groups that capture differences in interaction through the research design.

Prior to collecting data involving human subjects, sample recruitment, survey administration, and data collection was approved by the University of Massachusetts Lowell's Institutional Review Board. Then, I created an Upwork profile and developed a job specific to this research study. The job title clearly articulated that the job is for research purposes only and will involve completing the research survey instrument (Seeking Software Engineers for 15-minute Research Survey and Seeking Accountants for 15-minute Research Survey). The purpose and instructions were clearly communicated in the job posting and prior to participants completing the survey.

A total of 143 freelancers comprised the final sample after removal of nine disqualified cases, exceeding the estimated sample size needed for statistical power. Based on an a priori power analysis using G*Power, a widely used tool for conducting statistical power analyses, a sample size of 110 would provide appropriate statistical power to test the posed hypotheses (Faul et al., 2007; 2009). I conducted the power test based on a statistical power level of 0.95 for a linear multiple regression model with up to nine predictors including three primary predictor variables and a maximum of five

control variables ($p = 0.05$) to minimize threat of Type II error. The indicated sample size ranged from 77 to 110 based on the following assumptions of effect size.

“Cohen (1988) routinely suggested somewhat arbitrary guidelines for designation of small, medium, and large effect sizes for various hypothesis tests” (MacCallum, Browne & Sugawara, 1996: 138). Cohen’s (1988) overall standard for effect sizes is 0.10 (small), 0.30 (medium), and 0.50 (large), which does not accurately reflect the actual distribution of effect sizes in micro-oriented management research (Bosco et al., 2015), like my micro-level study. Therefore, an initial power analysis was conducted using a 0.35 effect size which, according to extant research, better reflects the average corrected effect size for this kind of research (Paterson et al., 2016). An effect size of 0.35 exceeds in magnitude about 80% of the effect sizes reported in the leadership and organizational behavior literatures (Paterson et al., 2016). Using this effect size resulted in a sample size of 77.

However, a smaller effect size is associated with interaction effects, like with hypothesis 3. Therefore, an additional power analysis was conducted using the more conservative effect size of 0.10, which indicated a sample size of 110 is needed to reach a statistical power level of 0.95. I employed an actual sample size exceeding this threshold ($N = 143$) while still remaining relatively close to the indicated power-test sample size to minimize Type I error associated with unduly large sample sizes. I continued to recruit participants until the indicated sample size was attained.

4.1.2. Measures

4.1.2.1. Boundary Spanning Scale

Ancona et al. (1992A) developed a boundary spanning scale that aids in understanding boundary spanning at a micro level. The six scouting items from this scale were used for this study because they measure high-frequency communicators, expertise, and access to information, which are characteristics of skilled digital freelancers. The six scouting items that were operationalized were: gathering information and resources, mapping, opening-up communication channels, scanning, and translating.

“I work to find out what competing organizations or groups (external to the contracting organization) are doing on similar projects” has been operationalized to “I share with the contracting organization what others in my network (e.g., other organizations) are doing on similar projects” and “I work to scan the environment inside and outside the contracting organization for marketing ideas and expertise” has been operationalized to “I share with the contracting organization what others in my network (e.g., other organizations) are doing for marketing ideas and expertise.” These measures of boundary spanning activities were used to determine the degree of scouting activities performed within the sample of skilled freelancers. Consistent with the validated scale, a 5-point Likert scale from “Strongly Disagree” (1) to “Strongly Agree” (5) was used. This operationalized scale can be found in Appendix A.

To allow for more insight into boundary spanning and to capture potential boundary spanning that may occur without a high degree of interaction (i.e., non-team centric items), the following open-ended questions were included: “Are there other ways you share your knowledge, expertise, or information with a hiring organization?” “If yes,

please describe in at least one paragraph.” The second question is open-ended and helped capture the less direct means of boundary spanning.

4.1.2.2. Interdependence

Task interdependence is the magnitude of which a task requires interaction. Interaction is when two or more entities react together and interdependence is when one entity cannot exist without the other, therefore, both concepts are related. In this study, interaction was measured using an established interdependence scale (Pearce et al., 1991) due to its reliability and validity of measuring employee-perceived, self-report task interdependence. In addition to collecting this self-report data, interaction was objectively measured by dummy coding the two freelancer groups through the purposive sampling. For hypothesis 1, interaction was included as an independent variable and was included as a control variable in hypotheses 2 and 3.

Some scale items go beyond interaction and focus on actual interdependence. For example, “My own performance is dependent on receiving accurate information from others” has been operationalized to “My work performance is often dependent on receiving accurate information from members of the contracting organization” which better gauges degree of interaction. Other scale items focus more on interaction, such as, “I work closely with others in doing my work” which has been operationalized to “I typically work closely with members of the contracting organization when doing my work.” Each scale item was answered using a 5-point Likert scale from “Strongly Disagree” (1) to “Strongly Agree” (5). The complete adapted scale can be found in Appendix B.

4.1.2.3. Expert Influence

A 6-item self-report expert power scale was used to capture subjective expert power. This included the 4-item self-report expert power scale as well as two additional items that were added to increase alpha as discussed further in the piloting measures section of this dissertation. The two added items were “I am often viewed by members of a contracting organization as having needed expertise for the job” and “I am often viewed by members of a contracting organization as unique knowledge for the job.”

The 4-item Hinkin et al. (1989) expert power scale was adapted from the seminal work of French et al. (1959). The items were operationalized for my study to determine freelancer’s perceived expert influence over others. This self-perspective rather than other-perspective approach to assessing subjective expert power is conceptually consistent with Anderson and Cappelli’s (2020) qualitative investigation of contractor-managers and deemed most relevant for the current research context that I am concerned with, which is the role of the freelancer’s believed expert influence on their own behaviors. For example, “My supervisor can give me good technical suggestions” was adapted for the present context as “I am often viewed by members of a contracting organization as having good technical suggestions to offer” and “My supervisor can share with me his/her considerable experience and/or training” was operationalized to “I am often viewed by members of a contracting organization as having considerable experience or training.” All items reflect expert power without a team or individual anchor and were used for the entire sample. Items were answered using a 5-point Likert scale from “Strongly Disagree” (1) to “Strongly Agree” (5). The complete, adapted 4-item expert influence scale can be found in Appendix C.

4.1.2.4. Strategic Orientation

For use in hypothesis 3, the survey included strategic orientation items (Soda et al., 2018) to gauge the moderating effect of collaborate on the relationship between expert power and boundary spanning. Collaboration was assessed using an established strategic orientation toward brokering scale (Soda et al., 2018). This scale has been used as a basis for emerging literature (e.g., Brennecke, 2020; Kwon et al., 2020) and includes six items that measure a collaborate strategic orientation. For example, “I forge connections between different people dealing with a particular issue” and “I introduce two people when I think they might benefit from becoming acquainted.” The strategic orientation scale used in my study included a preceding descriptive instruction and all items were answered using a 5-point Likert scale from “Strongly Disagree” (1) to “Strongly Agree” (5). The full scale (including both collaborate and arbitrage) and instruction can be found in Appendix D.

4.1.2.5. Referent Power

To further probe the perception of power from within the organization, social power items were included for exploratory purposes. Specifically, the 4-item referent power scale (Hinkin & Schriesheim, 1989) was applied as a post hoc test. The pre-existing items were operationalized for this research context. For example, “My supervisor can make me feel valued” was operationalized to “I am often viewed by members of a contracting organization as making them feel valued” and “My supervisor can make me feel like he/she approves of me” was operationalized to “I am often viewed by members of a contracting organization as approving of them.” Items were answered

using a 5-point Likert scale from “Strongly Disagree” (1) to “Strongly Agree” (5). This scale can be found in Appendix E.

4.1.2.6. Human Capital

Human capital measures primarily rely on the skilled freelancer’s profile and what others in the organization see. Education and experience are both traditional measures of human capital in the skilled digital freelance environment. Therefore, college degree and certifications completed were collected from the profile of each respondent to objectively gauge relative differences in human capital across individual participants. Additionally, for post hoc testing I also collected each freelancer’s resume, capturing work experience from their resumes as an additional recognized way to measure human capital (Kor & Sundaramurthy, 2009).

4.1.2.7. Control Variables

Each participant’s biographical information was collected for descriptive purposes and potential use as control variables based on a preliminary analysis. This information included age, gender, and ethnicity. Additionally, reputation rating was collected from the freelancer profiles as a potential control. Lastly, the scaled or categorical measure of interdependence was used as a control in hypotheses 2 and 3, as determined after review of preliminary results for each measure of interdependence.

A 3-item marker variable was included in the survey instrument to evaluate potential response bias and common method bias like the consistency in one’s response across items (Podsakoff et al., 2012; Simmering, et al., 2014; Miller & Simmering, 2022). The three items included in the survey to measure the marker variable were, “I think blue is a pretty color,” “I like the color blue,” and “Blue is a lovely color.” Each item was

rated on a 5-point Likert scale from “Strongly Disagree” (1) to “Strongly Agree” (5).

Lastly, an engagement check was included in the middle of the survey. The full survey instrument can be found in Appendix F.

4.1.2.8. Qualitative

I analyzed the open-ended boundary spanning questions utilizing best practices (Gioia, Corley, and Hamilton, 2013). First, I made a list of all the data inputs. Then, I read through each comment multiple times. Next, I carefully sorted each comment by common and repetitive words and phrases used, such as, trust, relationships, projects, and samples. Next, I took each comment and ordered them by themes based on commonalities. Four overarching themes emerged which were experience/insights, examples/projects, connections/relationships, and train/share. This means the additional ways that freelancers share their knowledge, expertise, or information with a hiring organization is by bringing their past experiences and own insights into the job, showcasing examples and projects of their work, demonstrating their connections and relationships in the relevant field and beyond, and indicating how they will train current internal employees as needed and share necessary information with them. This is discussed further in the results section.

4.1.3. Analytical Considerations and Approach

4.1.3.1. Piloting Measures

The preceding measures include validated scales that have been adapted for the study context. Thoroughly reviewing and piloting items are important steps in survey and scale development (Spector, 1992), and in this case were applied to adapted measures as

added degree of rigor. Prior to administering the main study survey, pilot and face validity tests were conducted for item refinement and finalization. Specifically, adapted survey measures were piloted at a public research university in the northeast. Following accepted convention, undergraduates were used for the pilot study (Ciuchta et al., 2018; Kronrod & Bart, 2018). Two hundred and thirty-six participants ($n = 236$) completed the survey.

I gathered preliminary feedback on scale reliability measures, whether questions were worded clearly, if there were any difficulties pertaining to survey completion, and on how long the survey took to complete. The 236-participant sample size well exceeded the recommendation that pilot studies encompass at least 10% of the final study's sample size (Lackey & Wingate, 1998). The pilot study sample size was larger than the main study due to convenience sampling through an established participant pool. Doing so provided a sample size that exceeded my target. I shared the survey with and invited comments and feedback from three academic practitioners at the University of Massachusetts Lowell who have over 50 years of combined research experience.

I provided participants with direct instructions through Qualtrics:

Answer the following items as if you were a skilled freelancer worker, "an individual who works on several projects at the same time or within a time frame, but for different organizations, and who provide a skilled service or labor (e.g., accounting, marketing, engineering), and not those who rent out assets (e.g., Airbnb), sell goods (e.g., Etsy), or perform unskilled labor (e.g., Uber driver)."

1. Please reflect on this definition.

2. Imagine you are a skilled freelancer working for multiple clients (e.g., organizations) through an online platform like Upwork. Upwork is an online platform for clients (e.g., organizations) in need of specific professional services (e.g., accounting, marketing, engineering) to connect with providers (e.g., skilled freelancers).

3. Please answer all the following items as you imagine the skilled freelancer would answer.

Following these instructions, participants were presented with the Qualtrics survey. Based on each validated scale, items were answered using a 5-point Likert scale from “Strongly Disagree” (1) to “Strongly Agree” (5). Once all the data was collected, it was thoroughly examined. Participants were removed if they did not complete the entire survey, failed the engagement check, or showed low engagement throughout the survey (e.g., selecting the same Likert number for each survey question).

Next, the data was analyzed using Stata. First, a correlation table analysis was conducted on all items which showed them all as being distinctive. Kaiser Meyer Olkin (KMO) test and Bartlett’s test are convenient options to measure sampling adequacy. Therefore, a factor test was done on all survey items which showed a significant Bartlett’s test indicating sufficient intercorrelations for conducting factor analysis. Kaiser Meyer Olkin (KMO) provides an overall measure of the overlap of shared variance between pairs of variables and should be at least 0.50 (Barrett & Morgan Jr, 2005). The KMO value for this study was considered adequate at 0.80.

The goal of this pilot study was to determine if the items for each measure were indeed separate dimensions and loaded as expected. This was done with Confirmatory

Factor Analysis (CFA). CFA is used to test the structure of scale instruments and to verify how well the variables being measured represent a construct (Anderson & Gerbing, 1988). There's a consensus in the literature that factors are retained when eigenvalues are above 1, that items are retained when magnitude loadings are above 0.40 (Hinkin, 1995), and that a Cronbach's alpha above 0.70 (Nunnally, 1978) indicates internal consistency (Klein, Astrachan, & Smyrnios, 2005). Results of the CFA yielded five factors with eigenvalues greater than 1. Items were retained if (i) they loaded 0.40 or more on a factor and (ii) did not load more than 0.40 on two factors. As expected, each scale loaded to its own factor for a total of five factors.

Task interdependence, expert power, boundary spanning, and referent power were the scales that were piloted. For task interdependence, the factor analysis showed a significant Barlett's test and good KMO value of 0.794. Scale reliability was also considered good with a scale reliability coefficient alpha of 0.805. The task interdependence scale contains three reverse coded items. Though the measure's overall alpha was sufficient, it made sense to unify the direction of the items based on debate in research on cognitive load of reverse-scored items. For example, reverse-scored items have been shown to reduce response validity (Schriesheim & Hill, 1981), introduce systematic error (Jackson, Wall, Martin & Davids, 1993), and lower item loadings when compared to positively worded items (Hinkin, 1995). Research has shown that once participants establish a pattern of responding to a survey, they may fail to adapt their responses to negatively worded items (Schmitt & Stults, 1986). Therefore, the reverse-scored task interdependence items were refined. For example, "I usually work

independently of members of the contracting organization” was operationalized to “I seldom work independently from members of the contracting organization.”

For the expert power scale, the factor analysis showed a significant Barlett’s test and good KMO value of 0.762. Scale reliability was acceptable with a scale reliability coefficient alpha of 0.745. Since alpha was below 0.80, two additional items were added to potentially bolster reliability in the main study. The two items are “I am often viewed by members of a contracting organization as having needed expertise for the job” and “I am often viewed by members of a contracting organization as having unique knowledge for the job.”

The boundary spanning factor analysis showed a significant Barlett’s test and good KMO value of 0.861. Scale reliability was also considered good with a coefficient alpha of 0.864. After further review, there were concerns pertaining to the focus of participants sharing information on what a competitor/other organization is doing, meaning, this could be viewed by the participant as an ethics or non-disclosure agreement breach, hence causing reluctance to respond. Therefore, “other/competing organization” wording was refined to reflect “network” wording due to non-disclosure agreement concerns pertaining to freelance professionals. For example, “I share with the contracting organization what other organizations are doing on similar projects” was operationalized to “I share with the contracting organization what others in my network (e.g., other organizations) are doing on similar projects.”

The referent power analysis showed a significant Barlett’s test and good KMO value of 0.782. Scale reliability was also considered good with a coefficient alpha of 0.884. No changes were made to this scale.

4.1.3.2. Common Method Bias

All research has the risk of response bias, or more generally, common method bias. Risks associated with common method bias were mitigated by applying the statistical strategy of the marker variable technique. The correlation between a marker variable and a substantive study variable is expected to be statistically equivalent to zero (Lindell & Whitney, 2001; Richardson, Simmering & Sturman, 2009; Williams, Hartman, & Cavazotte, 2010). An “ideal” a priori marker variable identified by past research is one’s attitudes towards the color blue (Miller & Chiodo, 2008; Simmering et al., 2015) which is considered conceptually uncorrelated with the current study’s primary variables, thus meeting the criteria for a marker-variable test.

Bias was further mitigated through the implementation of a rigorous research design which included the use of an engagement check, the use of established scales that have been validated in the literature, a credible sample collected from Upwork, a pilot test, and appropriate statistical analyses. Additionally, the survey included a distractor, specifically, a page break and instructions for completing succeeding sections of the survey. Once the data was collected, and before hypotheses testing, a careful review of the data was conducted, specifically regarding consistency in responses to survey items.

4.1.3.3. Statistical Approach

Analytical techniques for hypotheses testing included hierarchical multiple regression to isolate the variance explained in the variables of interest, mediation, and moderation. I did this to examine the contributions of predictors beyond previously entered predictors for statistical control purposes. For all regression models, I entered the block of control variables in the first step followed by substantive independent variables

in one or more additional steps to determine incremental variance explained to gauge whether the addition of variables improved the model's overall predictiveness, and to investigate moderation and mediation effects. I tested the curvilinear relationship by entering the squared term of the independent variable to determine the quadratic effect. For mediation, I used Haye's (2009) bootstrapping approach because "we should not be using tests that assume normality of the sampling distribution when competing tests are available that do not make this assumption and that are known to be more powerful" (Hayes, 2009: 411). Extant literature shows that bootstrapping is one of the more valid methods for testing mediating variable effects (MacKinnon et al., 2004; Williams & MacKinnon, 2008) and should be used as the method of choice. Lastly, for moderation, an interaction term was used between variables. Moderation is a method to determine whether a third variable influences the strength or direction of the relationship between the independent and dependent variable. For example, a moderator variable could change the strength of the relationship from strong to moderate, or to nothing at all. Results of the moderated regression were plotted at one standard deviation above and below the mean to interpret the interaction effect beyond its significance level.

Further, I used generally accepted analytical methods to conduct preliminary analyses on data assumptions and descriptive statistics. This included the assessment of data distributions, missing data, normality, and homoscedasticity. Additional analyses included correlations for the determination of multicollinearity, confirmation of internal validities for each multi-item measure, testing for common method bias, and robustness checks.

V. RESULTS

5.1. RESULTS

5.1.1. Descriptive Statistics

5.1.1.1. Preliminary Analyses

I collected data from 154 subjects for the main study. Before analyzing the data, I thoroughly reviewed it. Nine participants were removed due to failing the engagement check and two were removed for not including their names; this meant their rating could not be cross checked with their Upwork profiles. So, the final sample size was 143 ($N = 143$) skilled digital freelancers employed on the Upwork platform in the United States. Using Stata, I formed dummy, categorical, and ordinal variables for measures such as, gender, ethnicity, payroll, remote working time, and contract time, for example. For scaled measures, I formed variables utilizing the mean for each multi-item measure (boundary spanning, expert power, task interdependence, strategic orientation, and referent power) per respondent to conduct the analyses for hypothesis testing. Additionally, I cross checked each respondent's self-reported "reputation" rating (i.e., job success) with their Upwork profile. Each participant completed this accurately as their rating is visible to the public on their profile.

I designed the survey to force a response on each item before allowing participants to continue onto the next item. Participants were incentivized by payment, half of their hourly rate, upon completion and submission of their survey. This served to minimize and eliminate missing data in responses.

I ran descriptive statistics including all variables. Forty seven percent of participants identified as female ($n = 66$) and 53% as male ($n = 77$). See Figure 3.

Insert Figure 3 about here

Nine participants were born between 1946-1960 ($n = 9$), 26 between 1961-1976 ($n = 26$), 81 between 1977-1992 ($n = 81$), 27 between 1993-2008 ($n = 27$), and the average age of the participants was 47.5 years. See Figure 4.

Insert Figure 4 about here

Fifty-two percent of participants were accountants ($n = 74$), while 48% were software engineers ($n = 69$). See Figure 5.

Insert Figure 5 about here

For ethnicity, sixty one percent of participants identified as Caucasian ($n = 86$), 20% as Asian or Pacific Islander ($n = 29$), 8% as Hispanic or Latino ($n = 12$), 7% as African American ($n = 10$), and 4% as other ($n = 6$). See Figure 6.

Insert Figure 6 about here

Eighty percent of participants conduct remote work 100% of the time ($n = 114$) and 64% of participants have a 100% rating ($n = 91$). See Figure 7.

Insert Figure 7 about here

Regarding participant freelancer contracts, 36% last 1-3 months ($n = 51$), 22% last less than one month ($n = 32$), 16% last 4-6 months ($n = 23$), 14% last more than one year ($n = 20$), 8% last more than two years ($n = 12$), and 4% last 7-9 months ($n = 5$). See Figure 8.

Insert Figure 8 about here

Descriptive statistical results are shown in Table 1.

Insert Table 1 about here

I analyzed the data to determine if there were any outliers, and if the data and scales were normally distributed. Then, I determined skewness and kurtosis coefficients and ran histograms. Skewness measures the asymmetry of a distribution while kurtosis measures the heaviness of a distribution's tails relative to a normal distribution. Skewness and kurtosis were acceptable for all variables and visual examination of the histograms showed normal distributions; therefore, variables did not require transformations (Nadarajah, 2005). As noted previously, I included an engagement check in the survey instrument. Additionally, the correlation between the marker variable, the color blue, and the substantive variable of interest was low and statistically equivalent to zero. Therefore, I determined that common method bias was not a concern (Lindell et al., 2001). The correlation analysis can be found in Table 2.

Insert Table 2 about here

I ran a t-test analysis on task interdependence and boundary spanning for both groups (accountants and software engineers). Task interdependence $t = -1.828$, $df = 141$, $p = 0.069$; boundary spanning $t = -0.649$, $df = 141$, $p = 0.516$. The p-values are not significant in either test; therefore, the null hypothesis is not rejected for both tests. Given that there is not a statistically significant difference between the two groups on task interdependence and boundary spanning, I combined the two groups for all analyses.

I added the nine control variables to the regression analysis one at a time. Gender, rating, remote work, and contract time emerged as the most appropriate controls to use due to their low correlation with task interdependence, fit within the conceptual framework, being used as controls in prior research, and the benefit to statistical power in limiting control variables (Bernerth & Aguinis, 2016).

5.1.1.3. Construct Reliability

I followed similar steps to the pilot study analysis utilizing CFA. For task interdependence, the factor analysis showed a significant Barlett's test ($p\text{-value} < 0.05$), and good KMO value of 0.693. Scale reliability was considered acceptable with a scale reliability coefficient alpha of 0.691. To bolster scale reliability, I removed the item "I seldom plan my own work without needing to coordinate with members of the contracting organization," which increased alpha to 0.707. Upon reflection, this item represented a poor conceptual fit for the study context. Its removal does not alter the construct definition given the breadth of items remaining in the measure.

For the expert power scale, the factor analysis showed a significant Barlett's test ($p < 0.05$), and good KMO value of 0.724. The two items I added after the pilot study,

with the intention of increasing reliability by addressing content and construct validity, did indeed bolster reliability in the main study. Scale reliability coefficient alpha increased from 0.745 to 0.800. The boundary spanning factor analysis showed a significant Barlett's test ($p < 0.05$), and good KMO value of 0.844. Scale reliability was also considered good with a coefficient alpha of 0.858. Referent power factor analysis showed a significant Barlett's test ($p < 0.05$), and good KMO value of 0.896. Scale reliability was also considered good with a coefficient alpha of 0.897. I analyzed the strategic orientation scale with collaborate versus arbitrage items. For the strategic orientation collaborate scale, factor analysis showed a significant Barlett's test ($p < 0.05$), and good KMO value of 0.789. Scale reliability was acceptable with a coefficient alpha of 0.779. To further improve reliability, I removed the item "I will try to describe an issue in a way that will appeal to a diverse set of interests" which increased coefficient alpha to 0.824. Due to the scope of items remaining in this measure, the removal of that one item did not change the construct definition. For the strategic orientation arbitrage scale, factor analysis showed a significant Barlett's test ($p < 0.05$), and a low KMO value of 0.515. Scale reliability was also low with a coefficient alpha of 0.295 (Williams et al., 2010). Therefore, the arbitrage sub-scale was not included in the study (Soda et al., 2018).

Construct reliability results are shown in Table 3.

Insert Table 3 about here

5.1.2. Hypotheses

5.1.2.1. Hypothesis 1

I utilized quadratic regression analysis to test hypothesis 1, that skilled freelancer degree of interaction with internal organization members has a positive curvilinear relationship with their degree of boundary spanning behaviors performed in contracted roles. I tested hypothesis 1 in multiple ways to include robustness checks. Variance inflation factor (VIF) for all regressions are less than 10, so multicollinearity is not a significant issue (James et al., 2013). The baseline regression showed significance using combined groups; task interdependence has a strong positive relationship with boundary spanning ($\beta = 0.395, p = 0.009$). Time working remotely ($\beta = -0.086, p = 0.094$) and contract time ($\beta = -0.093, p = 0.061$) had a non-significant relationship with boundary spanning.

Next, I tested for the curvilinear relationship using the scaled interdependence measure and introduced the squared term to determine the quadratic effect. The regression indicated significance using combined groups; task interdependence has a strong positive effect on boundary spanning ($\beta = 0.388, p = 0.028$). Contract time ($\beta = -0.092, p = 0.085$) and time working remotely were not significant ($\beta = -0.085, p = 0.125$). Additionally, the squared interdependence measure was not significant ($\beta = -0.019, p = 0.915$). When I tested the model without the base linear term (i.e., non-squared term) there was no substantive change in result for the squared term ($\beta = -0.186, p = 0.264$). This shows support for a linear relationship, and not a curvilinear relationship. Therefore, hypothesis 1 is not supported regarding the curvilinear relationship, however, the finding still informs this study because it shows the significant linear relationship of task

interdependence with boundary spanning for the sample of skilled digital freelancers.

Hypothesis 1 results are shown in Table 4 with a visual in Figure 9.

Insert Table 4 about here

Insert Figure 9 about here

5.1.2.2. Hypothesis 2a

I used hierarchical multiple regression to determine the incremental variance accounted for by the predictor variable of interest in regards to the dependent variable, boundary spanning, after controlling for other potential influences. Skilled digital freelancer level of college education was used to measure human capital. When first analyzing human capital, I configured all five levels of education as an ordinal measure to capture a potential linear relationship between human capital and boundary spanning behaviors. Results were not significant ($\beta = 0.117, p = 0.334$), therefore, I further analyzed the relationship by contrasting low, medium, and high levels of education using ordinal variables. Low level of education = 1 and included no education or an associate degree; medium level of education = 2 and included a bachelor's degree; and high level of education = 3 and included a master's degree or PhD. Skilled digital freelancers with medium ($\beta = 0.452, p = 0.057$) and high ($\beta = 0.477, p = 0.056$) levels of education showed higher levels of boundary spanning behaviors, compared to those with low human capital (i.e., no college education or an associate degree). Results suggest marginal support of Hypothesis 2a; skilled freelancer human capital has a positive

relationship with their boundary spanning behaviors that is approaching statistical significance.

As construed in hypothesis 1, task interdependence is considered a necessary condition to the modeled relationships and was accordingly modeled as a control. Task interdependence was significant ($\beta = 0.357, p = 0.019$). This means the level of human capital among skilled digital freelancers contributes to explaining boundary spanning behaviors over and above freelancers' degree of interaction with others in their jobs. The results of these analyses are presented in Table 5 with a visual in Figure 10.

Insert Table 5 about here

Insert Figure 10 about here

Results remained robust when I included additional control variables that were reasoned to potentially influence degree of boundary spanning or its relationship with human capital. Additional controls that I included were gender, freelancer's reputation rating, remote work, and contract time. I entered the control variables in the first step, followed by independent variables in additional steps. This helped gauge whether the addition of variables improved the model's overall predictiveness.

I coded remote work as a dummy variable, meaning, freelancers who work remotely 100% of the time versus those who do not. Results demonstrated that skilled

digital freelancers who work remotely 100% of the time are not significantly lower on their level of boundary spanning behaviors ($\beta = -0.274, p = 0.097$).

I analyzed contract time by contrasting contract times of less than 6 months = 1, 6 – 12 months = 2, and greater than 1 year = 3. Contract time showed no significance. However, having a contract time of greater than 1 year had an unexpected negative relationship with a freelancer's degree of boundary spanning behaviors ($\beta = -0.388, p = 0.064$), though not reaching statistical significance. Gender and rating showed no significance.

5.1.2.3. Hypothesis 2b

I analyzed hypothesis 2b using Hayes (2009) bootstrapping approach which is considered one of the more valid methods for testing mediating effects (MacKinnon et al., 2004; Williams & MacKinnon, 2008). Results show that when using the linear measure of education level, human capital is not a factor in predicting expert power ($\beta = -0.074, p = 0.353$). Additionally, results demonstrate that expert power has no effect on boundary spanning behaviors ($\beta = -0.059, p = 0.700$). Therefore, hypothesis 2b is not supported; expert power is not a mediator of the relationship between human capital and boundary spanning behaviors. The results of these analyses are presented in Table 6 with a visual of the empirical model in Figure 11.

Insert Table 6 about here

Insert Figure 11 about here

I included the same control variables (task interdependence, gender, rating, remote work, and contract time) in the robustness checks. For path a of the mediation model, task interdependence was significant and had a positive effect ($\beta = 0.396, p = 0.001$) on a freelancer's degree of boundary spanning behaviors. For path b, task interdependence had a positive, significant effect on boundary spanning behaviors ($\beta = 0.396, p = 0.019$) and having a contract time of greater than 1 year had a negative effect that was approaching significance ($\beta = -0.410, p = 0.057$). For path c of the model, human capital was not significant ($\beta = 0.143, p = 0.238$). As I stated previously, expert power ($\beta = 0.024, p = 0.857$) had no effect on boundary spanning, therefore, the model was not supported.

5.1.2.4. Hypothesis 3

I used moderated mediation for hypothesis 3. This analysis utilized a bootstrapping approach (Hayes, 2009) and an interaction term between the variables of interest (expert power and collaborate strategic orientation). I mean centered these variables (moderator) for multicollinearity purposes. Freelancer's level of expert power had no significant effect on their boundary spanning behaviors ($\beta = -0.138, p = 0.399$), and the moderated relationship was also not significant ($\beta = 0.007, p = 0.969$). Therefore, results did not support hypothesis 3. The results of these analyses are presented in Table 7 with a visual of the empirical model in Figure 12.

 Insert Table 7 about here

Insert Figure 12 about here

When I conducted robustness checks, they demonstrated that the direct effects of a collaborate orientation remained significant with the controls in the model ($\beta = 0.316, p = 0.011$). Task interdependence had a positive, significant effect on boundary spanning ($\beta = 0.367, p = 0.041$) and having a contract time of over one year again had a negative but not statistically significant effect ($\beta = -0.399, p = 0.062$).

5.1.2.5. Post Hoc Tests

I further analyzed hypothesis 2a where working experience was used to measure human capital. I gathered working experience from each freelancer's resume. Results were not significant ($\beta = -0.001, p = 0.899$). This means that working experience as a measure of human capital does not relate to freelancer boundary spanning abilities. Regarding hypothesis 2b, working experience was also used as an additional test. Results suggest that working experience has a positive relationship with freelancer level of expert power ($\beta = 0.008, p = 0.040$). Task interdependence ($\beta = 0.389, p = 0.000$), rating ($\beta = -0.011, p = 0.000$), and working remotely ($\beta = 0.359, p = 0.004$) were all significant. I ran an additional post hoc analysis to further analyze expert power as a potential moderator, rather than a mediator, based on the conceptual possibility of expert power representing differences in freelancer behavior instead of a mediating form of influence. However, there was no support for this model. Results of these analyses are presented in Table 8.

Insert Table 8 about here

Post hoc tests I conducted for hypothesis 3 analyzed task interdependence and collaborate strategic orientation as interaction terms ($\beta = -0.262, p = 0.768$), as well as human capital and collaborate strategic orientation ($\beta = -0.576, p = 0.208$); both tests showed no support. I further probed modeled relationships in hypothesis 3, such as, whether a freelancer's level of collaborate orientation has a significant, positive relationship with their boundary spanning behaviors. In post-hoc analyses, I found a collaborate strategic orientation showed a directional relationship with boundary spanning behaviors. A freelancer's level of collaborate orientation has a significant, positive relationship with their degree of boundary spanning behaviors ($\beta = 0.319, p = 0.009$). This suggests that skilled digital freelancers who are strategically inclined to collaborate are also more likely to engage in boundary spanning behaviors.

5.1.2.6. Qualitative Insights

To gather a deeper understanding into freelancer boundary spanning, and to capture possible boundary spanning that may occur without a high degree of interaction (i.e., non-team centric items), I included open-ended questions in the survey. Eighty-eight freelancers completed this, however, 13 did not respond correctly (i.e., providing more information about their job, but not the boundary spanning aspects), so, the final sample of freelancers that were analyzed using this qualitative approach was seventy-five ($n = 75$). I reviewed the qualitative comments to glean further insights beyond the quantitative measurements and four themes emerged: experience/insights ($n = 27$), examples/projects ($n = 32$), connections/relationships ($n = 5$), and train/share ($n = 11$). This portrays the additional ways that freelancers share their knowledge, expertise, or information with a hiring organization. They bring their past experiences and personal

insights into the job, showcase examples and projects of their former work, demonstrate their connections and relationships in both the relevant field and beyond, and communicate how they are able to train internal employees and share necessary information with them. Qualitative insights are displayed in Figure 13.

Insert Figure 13 about here

Overall, the qualitative information showed that freelancers meet virtually, but directly with the hiring organization to showcase their expertise based on past experiences with clients, hence the theme experience/insights. For example, one freelancer wrote, “When applicable and not proprietary information, I try to share as much information and insight as possible with the contracting organization. This information comes from either previous experience, or knowledge/better practices gained from other projects.” Freelancer meetings are often conducted through phone calls, Zoom, or other online platforms like Slack and Microsoft Teams. During these meetings, their expertise is demonstrated through example work, mini projects, and websites, hence the theme examples/projects. For example, one freelancer stated, “I will often do a "mini" project to showcase my skills. Additionally, I have projects from trainings and other further learning opportunities that I use to display my knowledge and skillset.” Showcasing connections within and beyond the scope of the project was important for some freelancers, hence the theme connections/relationships. For example, one freelancer stated, “I often let my client know of other experts in my field that I can contact in the event I do not know the answer.” These are ways freelancer’s boundary span directly with the hiring organization and share their knowledge in a non-team centric way.

Freelancers also shared more about their boundary spanning behaviors in a more team centric way. This included building trust, transferring knowledge to current employees, adding value to the team, and sharing insights and skills, hence the theme train/share. For example, one freelancer said, “Before my contract ends, I train the employees on my knowledge as their success is key to my ongoing success.” Another freelancer stated, “Reviews from previous clients and certifications are pivotal in establishing trust. Especially in a freelancing environment.” I will further probe these qualitative implications in the discussion.

VI. DISCUSSIONS

6.1. DISCUSSIONS

6.1.1. Contributions

6.1.1.1 Theoretical Contributions

Current literature stresses the benefits of boundary spanning for the organization and intraorganizational members, and at times, for the individual boundary spanner, but is less focused on how someone becomes a boundary spanner (e.g., Tushman, 1977; Ancona, et al., 1990; Soda et al., 2018; Iorio, 2022). My study furthered boundary spanning theory by examining specific boundary spanning antecedents and behaviors of skilled digital freelancers, who are a potentially fluid and efficient way for organizations to boundary span, and represents an important step in connecting organizational boundary spanning concepts to this growing segment of the organizational workforce.

I highlighted the importance that task interdependence has regarding boundary spanning in a digital freelancer context. My study contributes to research in social embeddedness (Granovetter, 1973; Granovetter, 1985; Podolny & Baron, 1997; and Hayton, Carnabuci & Eisenberger, 2012), structural holes (Burt, 1992), and boundary spanning theories (Aldrich & Herker, 1977). Over a series of two primary data collections, a pilot, and main study, I established validity of the survey instrument to measure aspects of freelancers modeled herein as relevant to boundary spanning behaviors, such as, human capital and expert power. According to the knowledge

obtained from current literature, my study helps delve into boundary spanning theory regarding skilled digital freelancers.

Although theory surrounding social embeddedness is well established in the literature, its applications to boundary spanning have been limited. The lack of research using this perspective is perplexing given that social embeddedness is inherent to the boundary spanning process (Podolny et al., 1997; Schotter et al., 2017). In this study, boundary spanning is emphasized as the connection between a structural hole; a bridge between two disconnected nodes. That is, a relationship between freelancers and their networks. This involves exchange and interaction where they partner with their social ties to pursue goals and connect opportunities within and across boundaries, taking advantage of the many and often weak ties (as opposed to few strong ties) inherent to freelancers' networks. Weak ties, like the relationship between a freelancer and hiring organization, promote a diversity of connections, as well as access to a wide array of information (Granovetter, 1973). With this knowledge, freelancers can broker the exchange of information from their network to the organization to bridge structural holes (Granovetter, 1973; Burt, 1992).

In hypothesis 1 - Skilled freelancer degree of interaction with internal organization members has a positive curvilinear relationship with their degree of boundary spanning behaviors performed in contracted roles - freelancer task interdependence demonstrated a strong positive relationship with boundary spanning showing that freelancers who interact with others within their jobs engage in more boundary spanning behaviors compared to those who do not. Research states that freelancers often work as members of teams and with existing organizational employees

rather than in independent isolation (Brion et al., 2012); my study further contributes to this notion regarding task interdependence. The finding in my study for task interdependence suggests that interaction facilitates boundary spanning behaviors like knowledge sharing. Importantly, results suggest a linear relationship in which more interaction corresponds with more boundary spanning, rather than the posed curvilinear relationship in which some interaction is necessary but also sufficient.

Extending this finding to concepts of social embeddedness, I speculate that interaction may lay the groundwork for certain aspects of key relationships, such as trust, that support the ability for further boundary spanning. Social embeddedness theory indicates that the role of personal relationships, structures, and networks generate trust and relationships (Granovetter, 1973) and that individuals make decisions based on past interactions and repeated relationships (Granovetter, 1985; Podolny & Baron, 1997). Therefore, boundary spanning abilities may be associated with freelancer's social embeddedness. Freelancers who utilize their weak ties in ways that enhance social embeddedness (such as via repeated structured interactions even though these interactions lack the emotional intensity and reciprocity forming elements of strong ties) may be most able to engage in further boundary spanning behaviors. This informs both theory and practice by supporting the connection of structured or transactional task interdependence and boundary spanning while showcasing that freelancers coordinate, negotiate, and obtain knowledge from external stakeholders that is then – under these conditions of structured interdependence – shared with the hiring organization, recursively furthering their social embeddedness. This supports the proposed framework and the unique outside-inside approach of freelancers.

Theoretically, this dissertation demonstrates that knowledge sharing allows freelancers to bridge structural holes within the hiring organization's network while also aiding the organization to manage changes driven by external forces. This result also supports the outside-inside position of the freelancer, meaning, they can bridge gaps in knowledge, information, and skills that are lacking within the organization they are working for. This finding extends both social embeddedness theory and structural holes theory while applying an existing measure of individual boundary spanning behaviors to a unique, growing population of the current workforce, skilled digital freelancers.

Findings for hypothesis 2a suggest that a skilled freelancer's human capital as level of education marginally enhances their boundary spanning behaviors, meaning, their ability (i.e., level of college education) influences their degree of boundary spanning. This insight indicates that freelancers with a college degree can span more boundaries due to their additional level of knowledge gained through their education. Consequently, college level of education from a bachelor's to a master's degree were antecedents to freelancer's ability to contribute new information within their jobs. This phenomenon may be because those with higher levels of education are better equipped to act on organizational barriers by building relationships and collecting new knowledge (Droege et al., 2003). This finding supports the premise that freelancers with higher levels of human capital have an objective and perceived capability to contribute resources that the hiring organization's existing workforce does not possess, versus just contributing routine labor. However, working experience as human capital does not influence freelancer boundary spanning abilities. This may be due to education helping freelancers share knowledge learned with the hiring organizations.

Having a contract time of greater than 1 year had an unexpected, though not statistically significant, negative relationship with a freelancer's degree of boundary spanning behaviors portraying that a longer contract time corresponds to less boundary spanning behaviors. This pattern could be ignored due its lack of statistical significance but given the unexpected direction of the relationship, I reflected further on the possible cause. It could be because boundary spanning is helpful when starting a new project and is needed less as the project advances overtime (Kehoe et al., 2015).

Hypothesis 2b suggests that expert power does not mediate the relationship between human capital and boundary spanning. Therefore, human capital does not predict a freelancer's level of expert power. This result may be due to expert power being analyzed using the freelancer's subjective view and not the internal organizational employee view. However, hypothesis 2b does show support for working experience having an influence on a freelancer's level of expert power. This means a freelancer's working experience helps them to be viewed by internal organizational employees as having higher levels of expert power.

Working as part of a team benefits a freelancer's boundary spanning abilities. This could result in helpful outcomes for both the team and hiring organization such as being able to connect with new knowledge and bridge structural holes. There are also insights to be gleaned from results surrounding the control variables. For example, freelancers with contract times of greater than 1 year had a significant negative effect (that was approaching significance) on their degree of boundary spanning behaviors. This could be because boundary spanning is more helpful when starting a new freelance job and less helpful as the project time progresses and approaches an end. This suggests that

during the beginning of a project freelancers may span more boundaries and share those benefits with the hiring organization.

Working remotely 100% of the time did not significantly correspond to freelancer degree of boundary spanning behaviors. This result suggests that working remotely does not erode ability for boundary spanning. This is counterintuitive because remote work theoretically makes it more difficult to interact, build relationships, and, in turn, span boundaries. Additionally, trust or credibility is likely more difficult for a freelancer who works remotely 100% of the time, lacking in-person opportunities to connect and build relationships with organizational employees. Future research should explore the mechanisms through which freelancers compensate for these factors to engage in boundary spanning behaviors at a level similar to those in various formats of working environments, like a hybrid approach, and not fully remote. In terms of other tested controls, such as gender and reputation rating, there was no significance across hypotheses. This means gender and reputation rating have no effect on a freelancer's ability to span boundaries. These findings were interesting especially regarding reputation rating because it means there may be other factors beyond just a freelancer's rating that can influence their boundary spanning behaviors.

6.1.1.2. Signaling Expert Power

Extant literature demonstrates that expert power is based on employee's perception that a freelancer has specialized knowledge that the organization does not currently possess (Anderson et al., 2020). Results from hypothesis 2b showed that human capital is not a factor in predicting expert power. Freelancer expert power was not a mediator, meaning, human capital did not predict expert power and expert power did not

affect boundary spanning behaviors. This was tested in various ways to probe the hypothesis in depth, such as, human capital being tested as a categorical variable, as a range of years of education, as well as a sum of total years of education, and as an ordinal variable.

Existing theory shows that external experience gives contractor-managers an important source of expert power and an ability to influence organizational members (Anderson & Cappelli, 2021). Findings from this study suggest that a college degree may not signify a level of expert power. This may be because freelancers and organizational employees can both obtain college degrees. Perhaps a more unique differentiator and something specific to freelancers is needed to lead to an increase in expert power (i.e., how many hiring organizations the freelancer has previously worked for). Freelancers have shown samples of their work to showcase their skills. For example, one freelancer said, “I will often do a "mini" project to showcase my skills.” Doing so may be an alternative and stronger way to signal expertise and garner expert power. Though, even then it seems, based on the findings from my study, that freelancer use of expert power is not instrumental to their boundary spanning behaviors. Instead, given my study’s findings pertaining to structured interactions, it seems that freelancer exposure to internal employees and thus potential development of trust – a relational rather than transactional form of influence – facilitates boundary spanning behaviors. Therefore, relationships and who the freelancer is connected to may matter most.

6.1.1.3. Role of Collaboration in Boundary Spanning

Hypothesis 3 showed that a collaborative strategic orientation matters for boundary spanning. Freelancers with a collaborate strategic orientation had a significant,

positive effect on their boundary spanning behaviors demonstrating that those who are strategically inclined to collaborate have a higher boundary spanning ability. This is in line with task interdependence which also had a positive, significant effect on boundary spanning. Freelancers who work closely with organizational employees and collaborate, encompass more boundary spanning behaviors than those who do not collaborate.

A collaborate strategic orientation gives a freelancer the ability to influence internal organizational members (Soda et al., 2018). As current research shows, organizations struggle to build collaborative boundary spanning connections on which productivity, efficiency, and innovation depend (March, 1991; Ernst & Yip, 2009). Therefore, hiring skilled digital freelancers with a collaborative strategic orientation could aid hiring organizations.

Collaborative orientation is regulated by reputation rating based on what freelancers shared regarding the importance of reviews from previous clients. For example, one freelancer shared, “I often showcase my reviews from past clients and discuss my work history.” Freelancers may not be willing to share beyond their specific contracted organizational role due to lack of trust (Soda et al., 2018). A collaborating strategy connects disconnected others in a way that favors open and complete knowledge sharing among all involved parties (Soda et al., 2018).

Trust helps form a relationship that is open to knowledge sharing and information exchange and is often based on past experiences (Jarvenpaa & Shaw, 1998, Lipnack & Stamps, 2008). My study used trust and related power dynamics as theoretical underpinnings for freelancers’ strategic orientation towards their role with hiring organizations and showed that trust between the freelancer and organization helped the

freelancer collaborate. My study built on Soda et al., 2018, by demonstrating that freelancers who collaborate more effectively also span more boundaries such as with sharing knowledge. Trust aids in collaboration.

6.1.1.4. Suggestions for Future Research

Including and analyzing additional variables like trust may be beneficial to future research. Based on these research findings, expertise and influence through expert power may be less relevant to the freelancer context than trust, focusing more on relationships rather than credentials. Expertise could be more structurally determined whereas trust is less established in this context relative to non-freelance contexts. For example, training employees was a qualitative finding and could be part of the boundary spanning construct and the ultimate collaborative form of boundary spanning. Additionally, future studies should examine the level of agreement between the internal employees and organizational perspectives of freelancer's expert power as well as the self-reported perspectives of the freelancers.

Current research shows that star workers keep their knowledge for their advantage (Kehoe et al., 2015). The difference for freelancers appears to be the incentive and regulating mechanism of referrals and future chances for getting hired. As one freelancer said, "I give each project my all because I want to be referred to future jobs and clients." This supposition is a fruitful point for future research to explicitly examine, perhaps through experimental variation of regulating mechanisms among freelancers or even through implementation of comparable regulating mechanisms among star employees.

Freelancer boundary spanning abilities conceptually aligned with extant team-focused boundary spanning behaviors labeled as scouting in existing research. Scouting

abilities encompassed gathering information from the external environment to access information, knowledge, and expertise (Ancona et al., 1992). Therefore, the six scouting items were used to measure boundary spanning in my study. Through the qualitative data collected, some freelancers demonstrated performing tasks and showing projects, therefore, future research could consider utilizing additional boundary spanning dimensions from the scale, like ambassadorial and task coordinator items.

For instance, drawing on extant conceptual logic, ambassadorial activities tend to be more political, such as identifying threats in the organization and working to build support to combat the threats (Ancona et al., 1992). The qualitative data from my research provides reason to suggest that freelancers may engage in some ambassadorial activities, especially because of their unique position within an organization. For example, one freelancer stated, “The contracting organization sometimes requests me to educate their in-house developers on web portal administration, maintenance, and troubleshooting. I don't mind providing this support within the scope of current projects since I believe in customer and business success.”

The unique role boundary spanners play is associated with innovation and entrepreneurship because of a greater access to external partnering, critical resources, and information (Dodgson 1994; Ahuja, 2000). Boundary spanners are entrepreneurial, innovative (Challis, 1988), and “creative, lateral thinking rule-breakers who frequently combine a capacity for visionary thinking with an appetite for opportunism” (Leadbeater & Goss, 1998: 15). Extant literature implies that entrepreneurship and boundary spanning may be connected, but research at this point is scarce. Being new to the organization and

having a fresh outlook and entrepreneurial mindset could enable freelancers to better locate organizational threats that current internal employees may be overlooking.

Employees mostly build their relationships from work interactions that reinforce their preferences or with those who are similar (e.g., Byrne, 1969; ZTsui & O'Reilly). Carnabuci and Quintane (2023) found that employees with an innovative cognitive style benefit the least from brokerage positions, but their natural inclination is to build relationships across structural holes. Furthering this research by specifically studying freelancers could be beneficial because freelancers are contractual workers who bring a fresh perspective into a hiring organization while interacting with a variety of internal and external stakeholders.

Employees build networks that can harm their performance because they build relationships that supplement rather than complement their intellect (Carnabuci & Quintane, 2023). However, freelancers are different than traditional employees. They are unique individuals that build relationships, connections, and knowledge that aid their performance, intellect, and success. For example, one freelancer said, "I will share best practices related to modern recruiting tactics that have been shared with me via previous projects or via a third party." Another freelancer stated, "I often share information from another entity and past relationships; however, I seldom reveal the source of the information." This information shows freelancers using their past relationships and knowledge to further their working performance within new organizations. Therefore, future research could examine freelancers through an entrepreneurial lens to help advance extant literature (e.g., Carnabuci & Quintane, 2023) and further analyze qualitative data.

Regarding entrepreneurship, skilled digital freelancers are often viewed as encompassing entrepreneurial skills or being “digital entrepreneurs” (Nambisan, 2017) because they work for themselves. During the qualitative analysis of this study, several entrepreneurial elements were uncovered, such as, “streamlining processes to get the best results” and “sharing knowledge through creative outlets like blogging and YouTube.” Entrepreneurship could be an interesting and additional part of this model for future discoveries and could build upon and enrich existing entrepreneurship theories as well as boundary spanning theory regarding skilled digital freelancers. A recent study shows that social networks that bridge structural holes help adaptors and may hurt creators (Carnabuci & Quintane, 2023). In my study, the freelancer would be considered the creator. Therefore, studying freelance boundary spanning in this context could help contribute to the literature. Additionally, findings suggest that skilled digital freelancers who work remotely 100% of the time are significantly lower on their level of boundary spanning behaviors. This significant result suggests the need for additional future research regarding “digital entrepreneurs” (Nambisan, 2017).

Future research should examine the number of strong and weak ties regarding freelancers. This quantification would improve the ability to see if the numbers of their ties matter and to what degree. Additionally, examining structural network analysis would be beneficial regarding freelancer relationships and who they are connected to. Lastly, future studies may extend insights and generalizability by selecting vastly different sample groups. For example, accountants and marketers, or software engineers and lawyers. Choosing two groups who work differently, such as one group that primarily utilizes

computer programs versus one group that does not, could help compare skilled digital freelancer groups.

6.1.1.5. Managerial/Practical Implications

The transparency of freelancer reputation rating is important for freelancers to gain future job assignments. This acts as a supplementary mechanism and helps regulate collaboration when there is an absence of trust. Freelancer boundary spanning abilities can be utilized by organizations to share knowledge, expertise, information, and network connections. By leveraging these abilities, organizations can improve their decision-making, productivity, and competitiveness in their respective industries and bridge structural holes.

Findings from this study help further management literature by indicating that there are antecedents for skilled digital freelancers to serve as boundary spanners, like medium and high levels of education, allowing for the linking of internal networks with external sources of information (Aldrich & Herker, 1977). Education improves freelancer skills and knowledge, which can aid in their credibility with hiring organizations. This is consistent with current literature which shares a common human capital assumption; those with education tend to be more productive because of their ability to solve problems, adapt to shifting environments, and contribute to an organization's strategic goals (Becker & Huselid, 2006; Coff & Kryscynski, 2011). This study helps understand how individuals become boundary spanners, specifically in digital forms of jobs (e.g., Barley, Bechky & Milliken, 2017) which can aid organizations as they decide whether to hire freelancers.

My study contributes a better understanding of skilled digital freelancer human capital, expert power, and boundary spanning behaviors. This is helpful to freelancers so

they can understand what aspects of human capital affect their expert power and boundary spanning abilities. Since education as human capital does not support this in hypothesis 2b, another element of human capital may, like showcasing their skillset through example projects. Expert power focuses more specifically on knowledge and credentials, so examining freelancer projects and past work may be beneficial. Additionally, it would be helpful for future research to examine the perspectives of the internal employees and hiring organization on freelancer expert power. For example, a freelancer shared, “I often demonstrate my knowledge through past experiences and projects I have completed for other clients.” Therefore, studying expert power beyond the measurement scale could be beneficial.

Qualitative questions were implemented in the survey instrument with a focus on boundary spanning behaviors that could not be captured through the boundary spanning measurement scale. These questions were implemented to determine how else a freelancer may boundary span within their jobs, especially regarding non-team centric ways. This information showed that freelancers demonstrate their knowledge based on past experiences with past clients and that freelancers showcase their networks and relationships, as well as the value of those connections to the hiring organization. Additionally, freelancers shared the importance of building trust and transferring knowledge to current employees. This supports the notion that task interdependence is indeed important for boundary spanning. Freelancers mentioned that showcasing their knowledge through previous work and projects is imperative which connects to human capital, meaning, education through previous working experiences may matter more than having a college degree. These findings help understand freelancers in more depth, which

is helpful to the freelancers themselves, as well as the hiring organizations and employees they work with.

6.1.1.6. Limitations

The research was cross-sectional which cannot show causality, however, it is the first study analyzing this phenomenon. A longitudinal or experimental study would be needed to test causality. Two distinct freelance professions were chosen as a sampling strategy to capture a wide range or even distinct levels of task interdependence. However, when conducting preliminary analyses, there was no significant difference between the two sample groups of software engineers and accountants in level of task interdependence, therefore, sample group data was combined. Task interdependence was unexpectedly similar for both groups and had a positive effect on boundary spanning showing that freelancers who interact with organizational members during their jobs engage in more boundary spanning behaviors. In retrospect, this is likely due to the commonalities of work these two groups perform. For example, software engineers and accountants conduct most of their freelance jobs utilizing computer programs and working with hiring organizations in a remote, electronic manner. During the qualitative analysis freelancers often wrote about their work comprising of Zoom, Slack, and Microsoft Teams. Future studies should examine the perspective of the hiring organization (e.g., internal organizational employees) to both validate the perceptions expressed by freelancers and to uncover additional perspectives that contribute to a better understanding of the boundary spanning phenomena. The self-report measurement of expert power is a further limitation and area for future research to assess from the organizational perspective.

Despite no significant difference in substantive variables between the two sample groups, this two-group design did result in benefits. Benefits included generalizability of the study to a broader range of profession types, a larger sample size, and having a wider range in boundary spanning. Regarding generalizability, the two groups that were chosen were representative of skilled freelancer professions, and regarding broader categorization, these groups are generalizable due to being technology focused versus not.

6.1.1.7. Conclusion

Research on interdependencies that connect intra-organizational relationships with those that go beyond an organization's boundaries (Fuller et al., 2020; Altman et al., 2021) is emerging and growing. Boundary spanning has not been adequately studied on a micro-level because existing literature focuses on a macro-level and has not been able to completely inform theory and practice about whether skilled digital freelancers serve as boundary spanners, and, if so, what those antecedents may be and what form their boundary spanning behaviors may take. This dissertation examined the access, ability, and motive of skilled digital freelancers and contributed to a micro-level understanding of organizational boundary spanning in a social network theoretical context.

This study represents an initial step into understanding how these theories connect and relate to a unique group of individuals, skilled digital freelancers. Additionally, it provides a foundation for scholars to further develop and expand upon the proposed model. The hope is that this study stimulates future research that examines boundary spanning, structural holes, and social embeddedness theory to inform research and practice

surrounding a growing segment of the organizational workforce – skilled digital freelancers.

IX. LITERATURE CITED

9.1 LITERATURE CITED

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X. APPENDICES

10.1. APPENDIX A: Boundary Spanning Scouting Items

Measures of Boundary Spanning Scouting Activities (Ancona & Caldwell, 1992A)

- I share with the contracting organization what others in my network (e.g., other organizations) are doing on similar projects.
- I share with the contracting organization what others in my network (e.g., other organizations) are doing for marketing ideas and expertise.
- I share with the contracting organization what others in my network (e.g., other organizations) are doing for technical information and ideas.
- I share with the contracting organization what others in my network (e.g., other organizations) are doing as potential threats to their work.
- I share with the contracting organization what members need from external sources.
- I share with the contracting organization design problems that others in my network (e.g., other organizations) have.

10.2. APPENDIX B: Interdependence Items

Task Interdependence and Extrarole Behavior (Pearce & Gregersen, 1991)

- I typically work closely with members of the contracting organization when doing my work.
- I frequently coordinate my efforts with members of the contracting organization when doing my work.
- My work performance is often dependent on receiving accurate information from members of the contracting organization.
- The way I perform my skilled freelance job usually has a significant impact on members of the contracting organization.
- My work requires me to consult with members of the contracting organization quite frequently.
- I seldom work independently from members of the contracting organization.
- I seldom plan my own work without need to coordinate with members of the contracting organization.
- I often obtain information from members of the contracting organization to complete my work.
- To do my job, I usually spend most of my time talking to members of the contracting organization.
- In my job, I am frequently called on to provide information and advice to members of the contracting organization.

10.3. APPENDIX C: Expert Power Items

Expert Power Scale (Hinkin & Schriesheim, 1989)

- I am often viewed by members of a contracting organization as having good technical suggestions to offer.
- I am often viewed by members of a contracting organization as having considerable experience or training.
- I am often viewed by members of a contracting organization as having sound job-related advice to offer.
- I am often viewed by members of a contracting organization as having needed technical knowledge.
- I am often viewed by members of a contracting organization as having needed expertise for the job.
- I am often viewed by members of a contracting organization as unique knowledge for the job.

10.4. APPENDIX D: Strategic Orientation Items

Strategic Orientation toward Brokering Scale (Soda, Tortoriello, & Iorio, 2018)

Prompt: During your freelance assignments, when it comes to sharing your own specialized knowledge, information sources, or connections with the contracting organization, how well does each of the following statements describe your likely behavior or belief?

- I forge connections between different people dealing with a particular issue. (Collaborate)
- I introduce two people when I think they might benefit from becoming acquainted. (Collaborate)
- I believe meetings and open discussions are time consuming. (Arbitrage)
- I introduce people to each other who might have a common strategic work interest. (Collaborate)
- I point out the common ground shared by people who have different perspectives on an issue. (Collaborate)
- I see opportunities for partnerships between people. (Collaborate)
- If I believe it is not essential, I do not introduce people to each other. (Arbitrage)
- I will try to describe an issue in a way that will appeal to a diverse set of interests. (Collaborate)
- I like meeting people separately and recombining their insights on my own. (Arbitrage)

10.5. APPENDIX E: Referent Power Items

Referent Power Scale (Hinkin & Schriesheim, 1989)

- I am often viewed by members of a contracting organization as making them feel valued.
- I am often viewed by members of a contracting organization as approving of them.
- I am often viewed by members of a contracting organization as making them feel personally accepted.
- I am often viewed by members of a contracting organization as making them feel important.

10.6 APPENDIX F: Survey Instrument

Please answer the following items based on your current work as a skilled digital freelancer.

TASK INTERDEPENDENCE (Adapted from Pearce & Gregersen, 1991)

Likert Scale: 1 = Strongly Disagree; 2 = Disagree; 3 = Neutral; 4 = Agree; 5 = Strongly Agree	1	2	3	4	5
I typically work closely with members of the contracting organization when doing my work.					
I frequently coordinate my efforts with members of the contracting organization when doing my work.					
My work performance is often dependent on receiving accurate information from members of the contracting organization.					
The way I perform my skilled freelance job usually has a significant impact on members of the contracting organization.					
My work requires me to consult with members of the contracting organization quite frequently.					
I seldom work independently from members of the contracting organization.					
I seldom plan my own work without need to coordinate with members of the contracting organization.					
I often obtain information from members of the contracting organization to complete my work.					
To do my job, I usually spend most of my time talking to members of the contracting organization.					
In my job, I am frequently called on to provide information and advice to members of the contracting organization.					

EXPERT POWER (Adapted from Hinkin & Schriesheim, 1989)

Likert Scale: 1 = Strongly Disagree; 2 = Disagree; 3 = Neutral; 4 = Agree; 5 = Strongly Agree	1	2	3	4	5
I am often viewed by members of a contracting organization as having					
good technical suggestions to offer.					
considerable experience or training.					
sound job-related advice to offer.					
needed technical knowledge.					
needed expertise for this job.					
unique knowledge for the job.					
If you are engaged in this survey, select the number three.					

BOUNDARY SPANNING (Adapted from Ancona & Caldwell, 1992A)

Likert Scale: 1 = Strongly Disagree; 2 = Disagree; 3 = Neutral; 4 = Agree; 5 = Strongly Agree	1	2	3	4	5
I share with the contracting organization what others in my network (e.g., other organizations) are doing on similar projects.					
I share with the contracting organization what others in my network (e.g., other organizations) are doing for marketing ideas and expertise.					
I share with the contracting organization what others in my network (e.g., other organizations) are doing for technical information and ideas.					
I share with the contracting organization what others in my network (e.g., other organizations) are doing as potential threats to their work.					
I share with the contracting organization what members need from external sources.					
I share with the contracting organization design problems that others in my network (e.g., other organizations) have.					

Yes No

Are there other ways you share your knowledge, expertise, information, or network connections with a hiring organization?		
If yes, describe the other ways and how often. Please provide at least one paragraph.		

If not, describe the reasons or barriers keeping you from doing so. Please provide at least one paragraph.	
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MARKER VARIABLES (Miller & Simmering, 2022)

Likert Scale: 1 = Strongly Disagree; 2 = Disagree; 3 = Neutral; 4 = Agree; 5 = Strongly Agree	1	2	3	4	5
I think blue is a pretty color.					
I like the color blue.					
Blue is a lovely color.					

REFERENT POWER (Adapted from Hinkin & Schriesheim, 1989)

Likert Scale: 1 = Strongly Disagree; 2 = Disagree; 3 = Neutral; 4 = Agree; 5 = Strongly Agree	1	2	3	4	5
I am often viewed by members of a contracting organization as					
making them feel valued.					
approving of them.					
making them feel personally accepted.					
making them feel important.					

STRATEGIC ORENTIATION (Soda, Tortoriello, & Iorio, 2018)

Prompt: During your freelance assignments, when it comes to sharing your own specialized knowledge, information sources, or connections with the contracting organization, how well does each of the following statements describe your likely behavior or belief?

Likert Scale: 1 = Strongly Disagree; 2 = Disagree; 3 = Neutral; 4 = Agree; 5 = Strongly Agree	1	2	3	4	5
I forge connections between different people dealing with a particular issue.					
I introduce two people when I think they might benefit from becoming acquainted.					
I believe meetings and open discussions are time consuming.					
I introduce people to each other who might have a common strategic work interest.					
I point out the common ground shared by people who have different perspectives on an issue.					
I see opportunities for partnerships between people.					
If I believe it is not essential, I do not introduce people to each other.					

I will try to describe an issue in a way that will appeal to a diverse set of interests.					
I like meeting people separately and recombining their insights on my own.					

DEMOGRAPHICS

- 1.) What is your birth year? _____
- 2.) The gender you identify with is:
 - a. Male
 - b. Female
 - c. Non-binary/third gender
 - d. Prefer not to say
- 3.) Which ethnicity best describes you? (Please choose only one.)
 - a. Hispanic or Latino
 - b. Asian or Pacific Islander
 - c. Caucasian
 - d. African American
 - e. Other _____
- 4.) What is your reputation rating? _____
- 5.) Do you hold a payroll job in addition to your freelance Upwork work?
 - a. Yes, part time
 - b. Yes, full time
 - c. No
- 6.) (if yes to Question 5) What is your current or most recent professional working title? _____
- 7.) What percentage of your weekly freelance work is typically conducted remotely (rather than at the employing organization's site)?
 - a. 0%
 - b. Up to 25%
 - c. Up to 50%
 - d. Up to 75%
 - e. 100%

- 8.) What is your average contract time (in months) with an organization?
- a. < 1 month
 - b. 1 – 3 months
 - c. 4 – 6 months
 - d. 7 – 9 months
 - e. 10 – 12 months
 - f. > 1 year
 - g. > 2 years
- 9.) On average, how often do you obtain repeat contracts with the same organization(s) you are working for through Upwork?
- a. 0% of the time
 - b. Up to 25% of the time
 - c. Up to 50% of the time
 - d. Up to 75% of the time
 - e. 100% of the time
- 10.) On average, how often do you sign a Non-disclosure Agreement (NDA) with the organizations you are working for through Upwork?
- f. 0% of the time
 - g. Up to 25% of the time
 - h. Up to 50% of the time
 - i. Up to 75% of the time
 - j. 100% of the time
- 11.) Please provide your name so we can confirm completion for payment.
- 12.) Any questions or clarifying comments to add? _____

XI. TABLES

11.1 TABLE 1: Descriptive Statistics

		Obs.	%	Mean	Std. Dev.	Range
Age	1946-1960	9	6			
	1961-1976	26	18			
	1977-1992	81	57			
	1993-2008	27	19			
Ethnicity	Caucasian	86	61			
	Asian or Pacific Islander	29	20			
	Hispanic or Latino	12	8			
	African American	10	7			
	Other	6	4			
Gender	Male	77	53			
	Female	66	47			
Occupation	Accountant	74	52			
	Software Engineer	69	48			
Boundary Spanning		143	100	2.650	0.910	4
Contract Time	Less than 1 month	32	22	1.489	0.812	2
	1-3 months	51	36			
	4-6 months	23	16			
	7-9 months	5	4			
	More than 1 year	20	14			
	More than 2 years	12	8			
Expert Power		143	100	4.307	0.575	2.25
Human Capital	Low level of education (no education or an associate degree)	11	8	2.314	0.610	2
	Medium level of education (a bachelor's degree)	78	54			
	High level of education (a master's degree or PhD)	54	38			
Human Capital	Working experience	143	100	12.028	9.795	45
Remote Work	Less than 100%	29	20	4.279	1.460	4
	100%	114	80			
Strategic Orientation		143	100	3.527	0.705	3.7
Task Interdependence		143	100	3.717	0.511	3

11.2 TABLE 2: Correlation

Variable	1	2	3	4	5	6	7	8	9	10	11
1. Boundary Spanning	1.000										
2. Task Interdependence	0.183*	1.000									
3. Expert Power	0.015	0.379*	1.000								
4. Human Capital	0.102	0.038	-0.087	1.000							
5. Referent Power	0.118	0.222*	0.309*	-0.031	1.000						
6. Strategic Orientation - Collaboration	0.268	0.144	0.198*	-0.017	0.191	1.000					
7. Marker Variable	0.155	0.143	0.115	0.030	0.235	0.081	1.000				
8. Gender	0.066	-0.023	-0.066	0.004	0.003	0.123	0.006	1.000			
9. Rating	0.001	-0.118	-0.191*	-0.043	-0.123	-0.123	-0.113	0.076	1.000		
10. Remote Work	-0.127	0.042	0.247*	-0.083	0.075	-0.128	0.126	0.042	0.137	1.000	
11. Contract Time	-0.131	0.180*	0.044	0.002	0.003	-0.031	0.027	-0.259*	0.154	-0.031	1.000
*Correlation is significant at the 0.05 level (2-tailed)											

11.3 TABLE 3: Construct Reliability

	Kaiser-Meyer-Olkin Measure of Sampling Adequacy	Alpha
Boundary Spanning	0.844	0.859
Expert Power	0.724	0.800
Referent Power	0.896	0.897
Strategic Orientation, Collaborate	0.789	0.825
Task Interdependence	0.693	0.707

11.4 TABLE 4: Hypothesis 1

	Model 1	Model 2	Model 3
Gender	0.052	0.053	0.075
Rating	0.003	0.003*	0.003
Remote Work	-0.086	-0.085	-0.074
Contract Time	-0.093	-0.092	-0.072
<i>Task Interdependence</i>	0.395*	0.388*	
<i>Task Interdependence Squared</i>		-0.019	-0.186
R ²	0.081	0.081	0.043
*p < 0.05			

11.5 TABLE 5: Hypothesis 2a

	Model 1	Model 2
Task Interdependence	0.378*	0.357*
Gender	0.033	0.047
Rating	0.002	0.002
Remote Work	-0.276	-0.274
Contract Time	-0.185	
Contract Time of less than 6-12 months		0.154
Contract Time of greater than 1 year		-0.388
<i>Human Capital</i>	0.117	
<i>Medium level of education (a bachelor's degree)</i>		0.452
<i>High level of education (a master's degree or PhD)</i>		0.477
R ²	0.084	0.104
*p < 0.05		

11.6 TABLE 6: Hypothesis 2b

	Path A (human capital -> expert power)	Path B (expert power -> boundary spanning)	Path C (human capital -> boundary spanning)
Task Interdependence	0.396*	0.396*	0.364*
Gender	-0.047	0.049	0.048
Rating	-0.012*	0.001	0.003
Remote Work	0.335*	-0.259	-0.256
Contract Time			
Contract Time of less than 6-12 months	0.071	0.104	0.150
Contract Time of greater than 1 year	-0.044	-0.411	0.407
<i>Human Capital (linear measure of education level)</i>	-0.074		0.143
<i>Expert Power</i>		-0.059	
R ²	0.228	0.087	0.095
*p < 0.05			

11.7 TABLE 7: Hypothesis 3

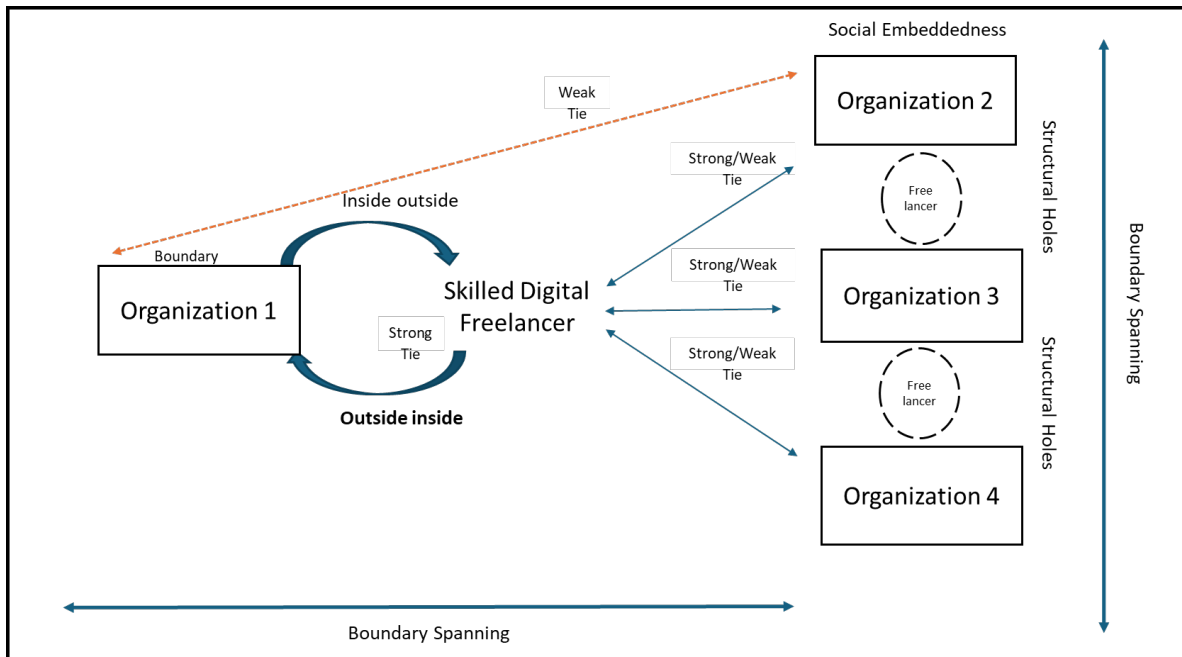
	Path A (human capital -> expert power)	Path B (moderated mediation)	Path C (human capital -> boundary spanning)
Task Interdependence	0.396*	0.367*	0.364*
Gender	-0.047	-0.012	0.048
Rating	-0.012*	0.004	0.003
Remote Work	0.335*	-0.167	-0.256
Contract Time			
Contract Time of less than 6 months			
Contract Time of less than 6-12 months	0.071	0.091	0.150
Contract Time of greater than 1 year	-0.044	-0.399	0.407
Human Capital (linear measure of education level)	-0.074		0.143
Expert Power		-0.138	
Strategic Orientation		0.316*	
Expert Power x Strategic Orientation		0.007	
R ²	0.228	0.142	0.095
*p < 0.05			

11.8 TABLE 8: Post Hoc Analysis

	Path A (human capital) -> expert power)	Path B (expert power -> boundary spanning)	Path C (human capital) -> boundary spanning)
Task Interdependence	0.389*	0.396*	0.373*
Gender	✓ -0.054	0.049	0.052
Rating	-0.011*	0.001	0.002
Remote Work	0.359*	✓ -0.259	✓ -0.281
Contract Time			
Contract Time of less than 6-12 months	0.068	0.104	0.101
Contract Time of greater than 1 year	✓ -0.067	✓ -0.410	✓ -0.405
Human Capital (working experience from resume)	0.008*		✓ -0.001
Expert Power		✓ -0.059	
R ²	0.244	0.087	
*p < 0.05			

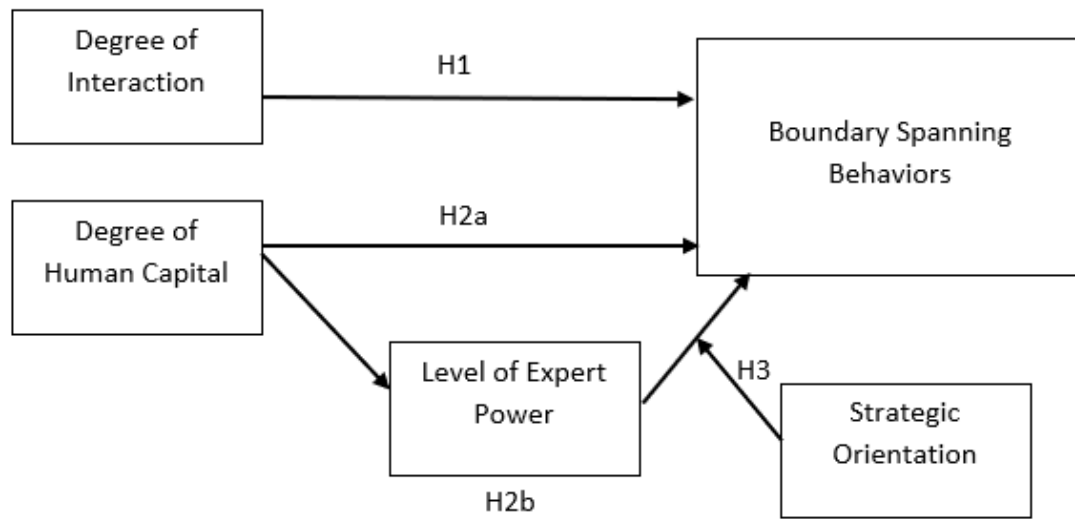
XII. FIGURES

12.1. FIGURE 1: Integrated Conceptual Depiction of Freelancers

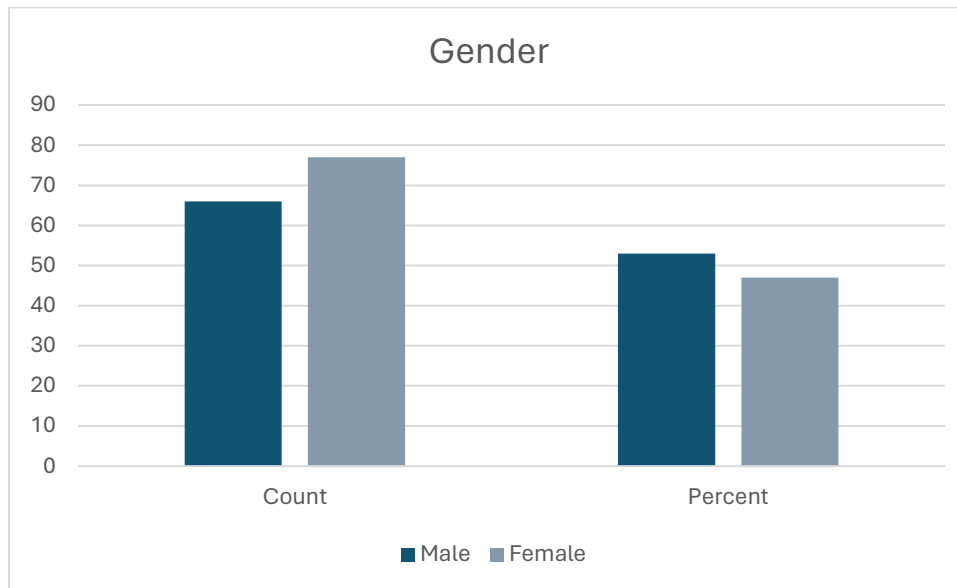


6

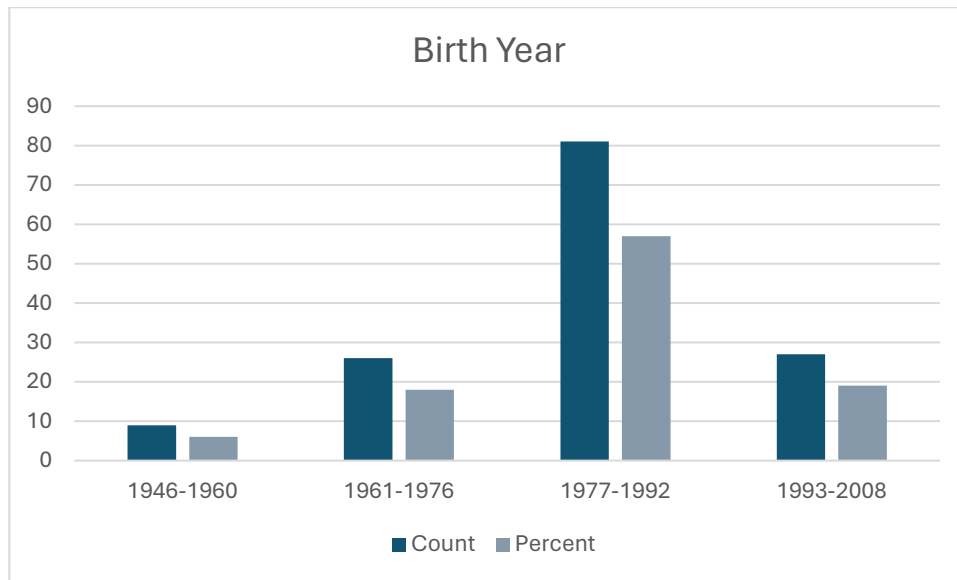
12.2. FIGURE 2: Empirical Model



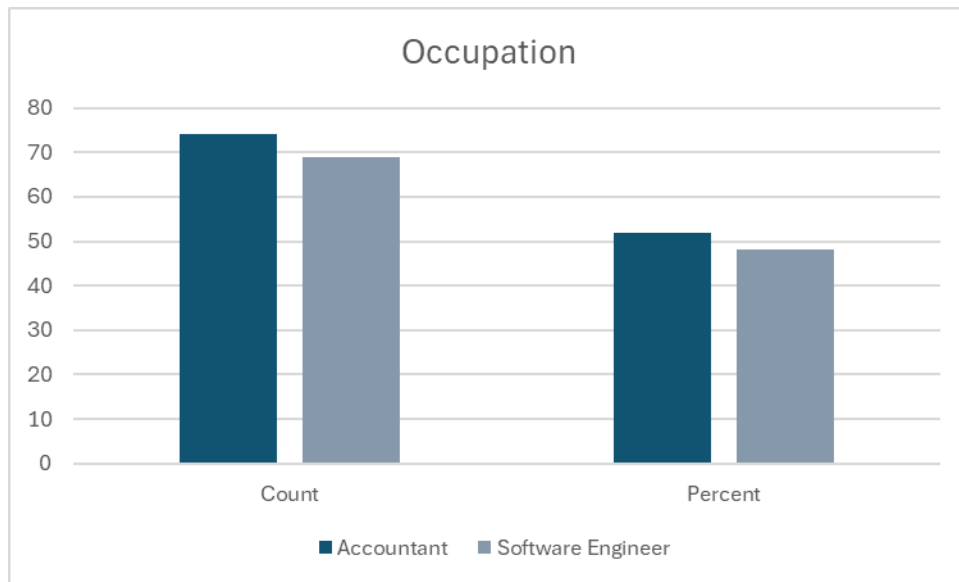
12.3. FIGURE 3: Gender



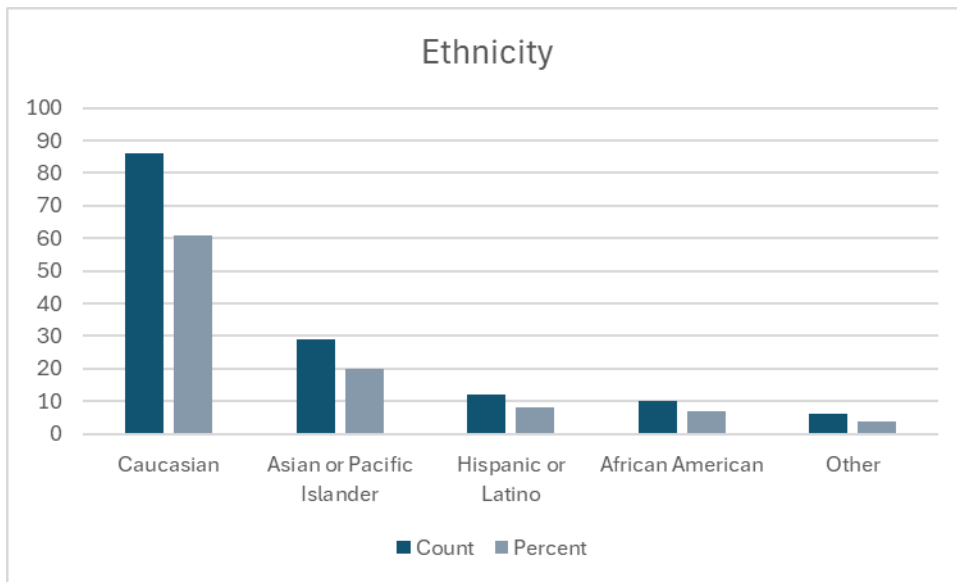
12.4. FIGURE 4: Age



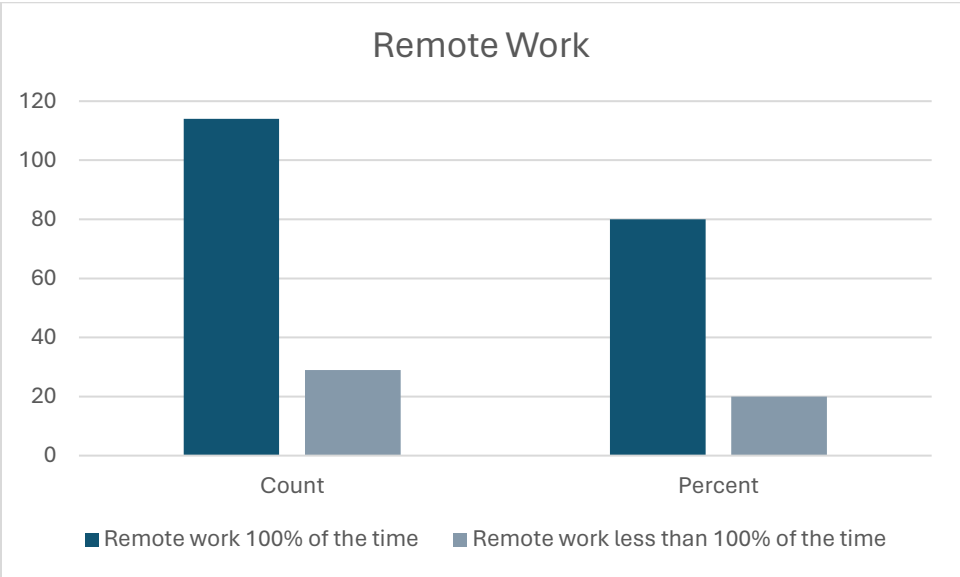
12.5. FIGURE 5: Occupation



12.6. FIGURE 6: Ethnicity



12.7. FIGURE 7: Remote Work



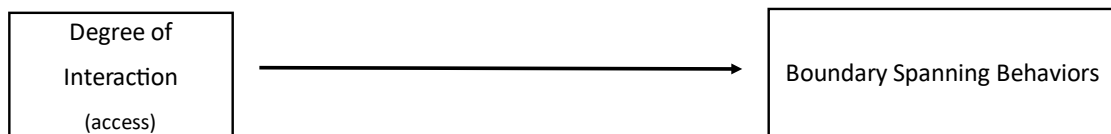
12.8. FIGURE 8: Contracts



12.9. FIGURE 9: Visual of Hypothesis 1 Results

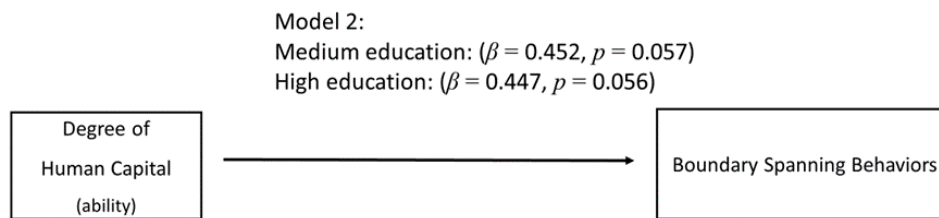
Supports a linear relationship: ($\beta = 0.395, p = 0.009$)

Does not support a curvilinear relationship: ($\beta = -0.019, p = 0.915$)



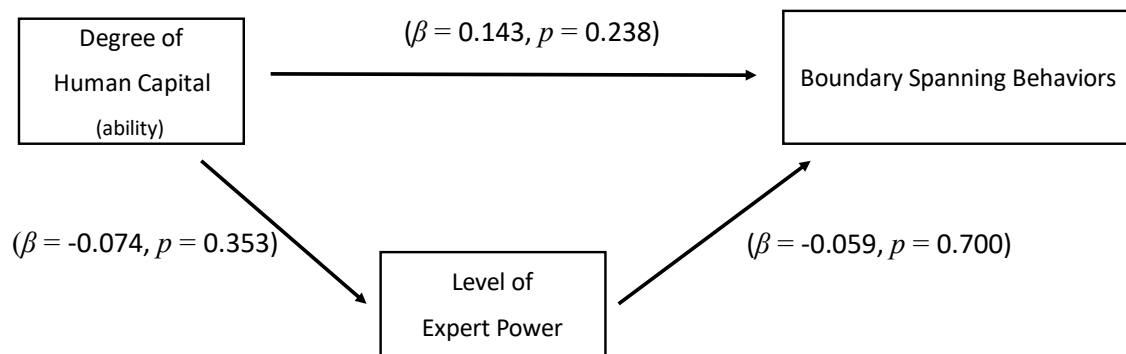
H1: Skilled freelancer degree of interaction with internal organization members has a positive curvilinear relationship with their degree of boundary spanning behaviors performed in contracted roles.

12.10. FIGURE 10: Visual of Hypothesis H2a Results



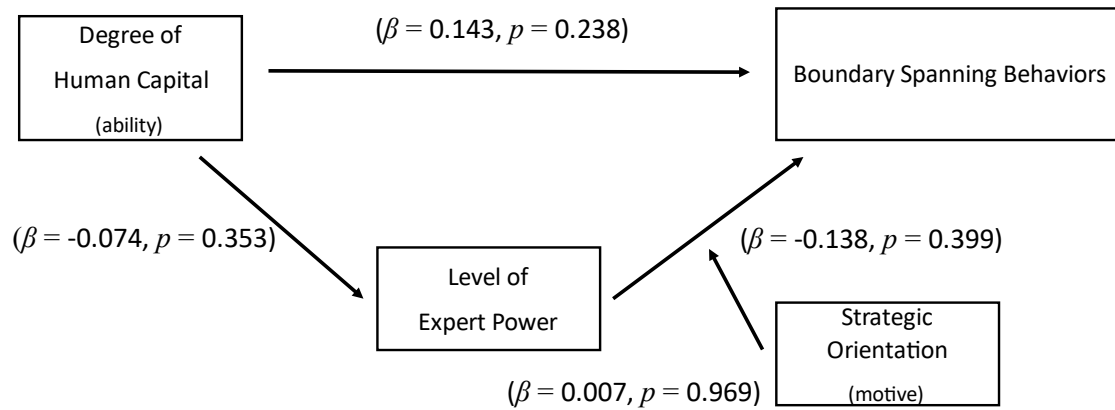
H2a: Skilled freelancer degree of human capital has a positive relationship with their degree of boundary spanning behaviors performed in contracted roles.

12.11. FIGURE 11: Visual of Hypothesis H2b Results



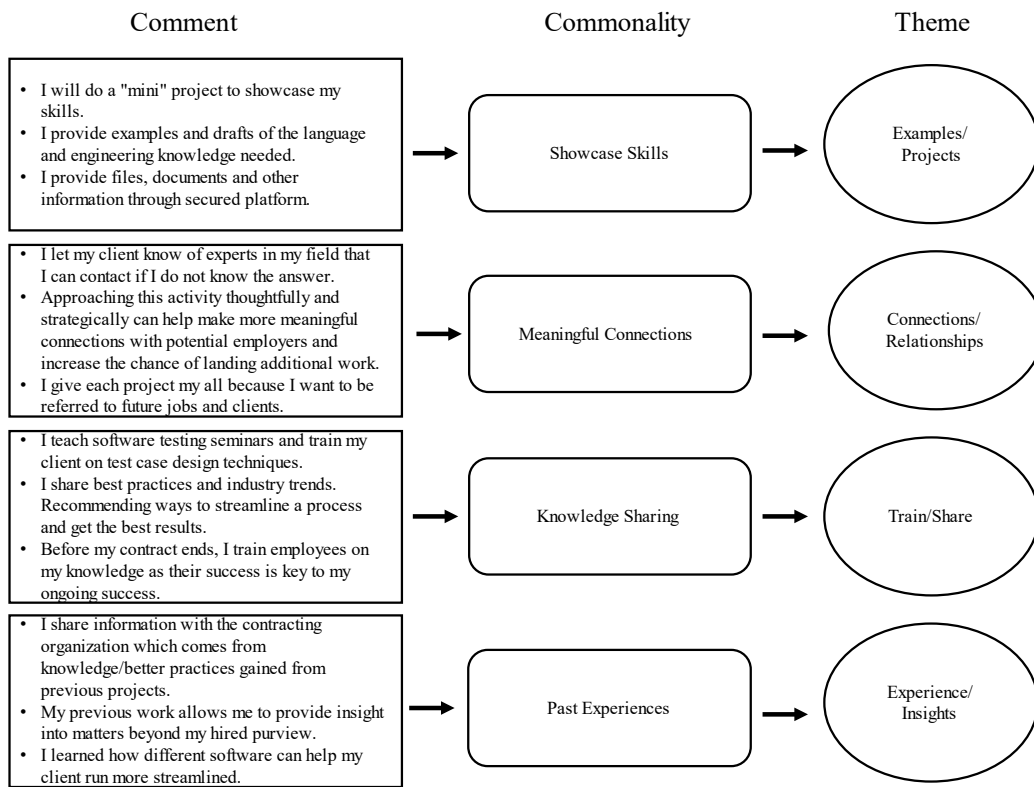
H2b: The relationship between skilled freelancer degree of human capital and degree of boundary spanning behaviors is mediated by their level of expert power.

12.12. FIGURE 12: Visual of Hypothesis H3 Results



H3: Freelancer collaborate orientation amplifies the positive relationship between freelancer expert power and their degree of boundary spanning behaviors.

12.13. FIGURE 13: Qualitative Process



XIII. BIOGRAPHICAL SKETCH OF AUTHOR

13.1. Biographical Sketch of Author

Holly J. Lalos is a recent PhD graduate in Management (Leadership and Organization Studies) from the Manning School of Business at the University of Massachusetts Lowell. She earned her M.S. in Innovation and Technological Entrepreneurship and her B.S. in business finance and marketing from the University of Massachusetts Lowell. She currently serves as director of the Rist DifferenceMaker Institute and adjunct faculty member in the Manning School of Business at the University of Massachusetts Lowell. She has spent more than a decade in a variety of leadership positions. Her current responsibilities involve providing management, leadership, and guidance to entrepreneurship ecosystem project staff, along with faculty and students participating in the university's multiple entrepreneurial endeavors. Holly has achieved numerous awards at the Deshpande Symposium on Innovation and Entrepreneurship in Higher Education, the Global Consortium of Entrepreneurship Centers, the United States Association for Small Business and Entrepreneurship, and the VentureWell Open Conference, and received numerous grants from M&T Bank, Eastern Bank, and VentureWell. Her research interests include leadership, boundary spanning, entrepreneurship, gig-work, and organizational change. She recently presented, "Who are Boundary Spanning Leaders?" at the Eastern Academy Management Conference.

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